Valkyrie-4 (VOS Model V4-LI800): Single-Launch Deployment and In-Plane Phasing for Rapid-Revisit Imaging Over India Using STK Astrogator

MECH 518: Orbital Mechanics — Final Project Report

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Abstract

Background: Rapid-revisit observation within a low-inclination band is motivated by the concentration of numerous time-critical threats (wildfires, volcanic ash, coastal flooding, severe storms) and high-value logistics activity at low latitudes.

Method: This project uses STK Astrogator to design *Valkyrie-4*, a constellation of four satellites launched from a single upper stage. At $i = 28.5°$, the vehicle injects the stack into a 200 km circular parking orbit. All four satellites separate after the upper stage completes a two-burn rise to a 300 km circular deployment orbit. Phasing is accomplished by timing: each satellite performs a finite-burn insertion to a shared 800 km circular operational orbit after loitering for a varying number of rotations in the deployment orbit.

Results: An upper-stage total ΔV of $60.534\,\mathrm{m\,s^{-1}}$ is reported by STK Astrogator. With $\Delta V \approx 266\,\mathrm{m\,s^{-1}}$ per satellite and ~ 764 s burn duration, satellite insertions are finite burns. The orbital period of 800 km is roughly 6050 s. Valk1: 1, Valk2: 3, Valk3: 6, and Valk4: 9 revolutions were the loiter counts employed.

Conclusion: Valkyrie-4 uses Astrogator targeting and repeat-count logic with clear ΔV accounting and evidence via MCS segment summaries to demonstrate a clean single-launch constellation deployment where along-track spacing is predominantly achieved by *timing* rather than expensive plane adjustments.

Keywords: STK Astrogator, constellation deployment, LEO, in-plane phasing, true anomaly, mission control sequence

1 Introduction

1.1 Company scenario and business case

mission framing (business case). A four-satellite low-inclination LEO imaging constellation called *Valkyrie-4* is being developed by *Valkyrie Orbital Solutions (VOS)* to provide quick situational awareness throughout the low-latitude belt. Our main client AOI for this final project concept is **India**, with a focus on infrastructure monitoring and disaster response in both interior and coastal areas. With one launch, one upper-stage delivery, and deterministic in-plane phasing that can be duplicated for bigger future constellations, the device is marketed as a straightforward, scalable "pathfinder" constellation.

Customers who pay for time-sensitive geospatial intelligence over India, such as emergency management agencies (flooding/cyclones), agriculture and water-resource stakeholders (drought and crop stress), coastal

authorities (storm surge and shoreline change), and logistics/energy operators who require quick awareness during disruptions, can purchase Valkyrie-4 as a service (subscription + surge-response tasking). Instead of rigorous local solar time control, the mission design promotes **fast revisit** inside the target latitude band that frequently crosses the India AOI.

Customer pain point: Customers frequently require *timely* imaging more than ultra-fine resolution in disaster response and infrastructure monitoring. Excellent photography can be obtained with a single spacecraft, but it has revisit gaps; many planes address this issue but complicate launch and operations. Valkyrie-4 is positioned by VIS as a "fast-to-field" capability with a single launch, just one plane, a low operational burden, and consistent time-staggered overpasses.

Product concept: Customers buy:

- **Routine monitoring (subscription) over India:** recurring analytics and planned imaging for a fixed AOI polygon (in this case, India AOI). In addition to standardized map deliverables like time-series imagery, water-extent tracking for major river basins, coastline change indicators, vegetation/crop-stress proxies, and infrastructure-change markers near critical corridors, customers can choose a cadence (e.g., daily/weekly).
- **Surge tasking (event response) in India:** high-priority collections during significant marine incidents, landslides, wildfire smoke occurrences, cyclones on the Arabian Sea and Bay of Bengal coasts, and monsoon flooding. Valkyrie-4 prioritizes passes while upholding imaging restrictions and safe operations once the client submits an AOI + time frame + priority. Short incident briefs and quick before/after comparisons are among the items to be delivered.
- **API delivery (integration-ready):** products that can be directly integrated into GIS and operations dashboards. GeoTIFF images, information (timestamp, viewing geometry, quality flags), and optional change masks for India AOI regions are among the outputs. Latency and revisit targets establish service tiers.

1.2 Mission objectives and technical approach

In order to illustrate (i) staged deployment, (ii) in-plane transfers, and (iii) true-anomaly phasing through controlled loiter timing, this paper designs and simulates *Valkyrie-4* as a single-plane constellation. In order to emphasize propellant-efficient and operationally straightforward operations, the design purposefully eliminates plane transitions (no RAAN phasing).

The mission sequence is:

- Launch to a 200 km circular parking orbit at $i = 28.5°$.
- Upper stage performs a two-burn transfer to a 300 km circular deployment orbit and releases four satellites.
- Each satellite loiters for a prescribed number of revolutions, then performs a finite-burn insertion to a common 800 km circular operational orbit.

Deliverables in this report: STK visual outputs, satellite design detail table, orbit-by-phase table, mission ΔV and loiter needs table, mission chronology, and Astrogator MCS evidence (figures + attached MCS text summaries).

2 Sales pitch: Rapid-response monitoring over India (AOI)

From the standpoint of the client, Valkyrie-4 is a low-latitude belt rapid-response imaging service, with **India** as the main region of interest. This is the value proposition:

- **Fast revisit over India:** The possibility of frequent observations over the India AOI during a 24-hour timeframe is increased when four spacecraft are phased in one plane to spread overpasses over time.
- **Simple operations:** solely in-plane motions, lowering expenses and operating risk (no RAAN modifications).
- **Scalable logic:** Naturally, the loiter-then-insert strategy may be applied to more satellites (and eventually to several planes).
- **Clear ΔV ownership:**Satellites pay insertion to the final orbit after the upper stage delivers to the deployment orbit.

2.1 Product view (STK visualization)

Figure 1 is the customer-facing visualization of the delivered operational configuration.

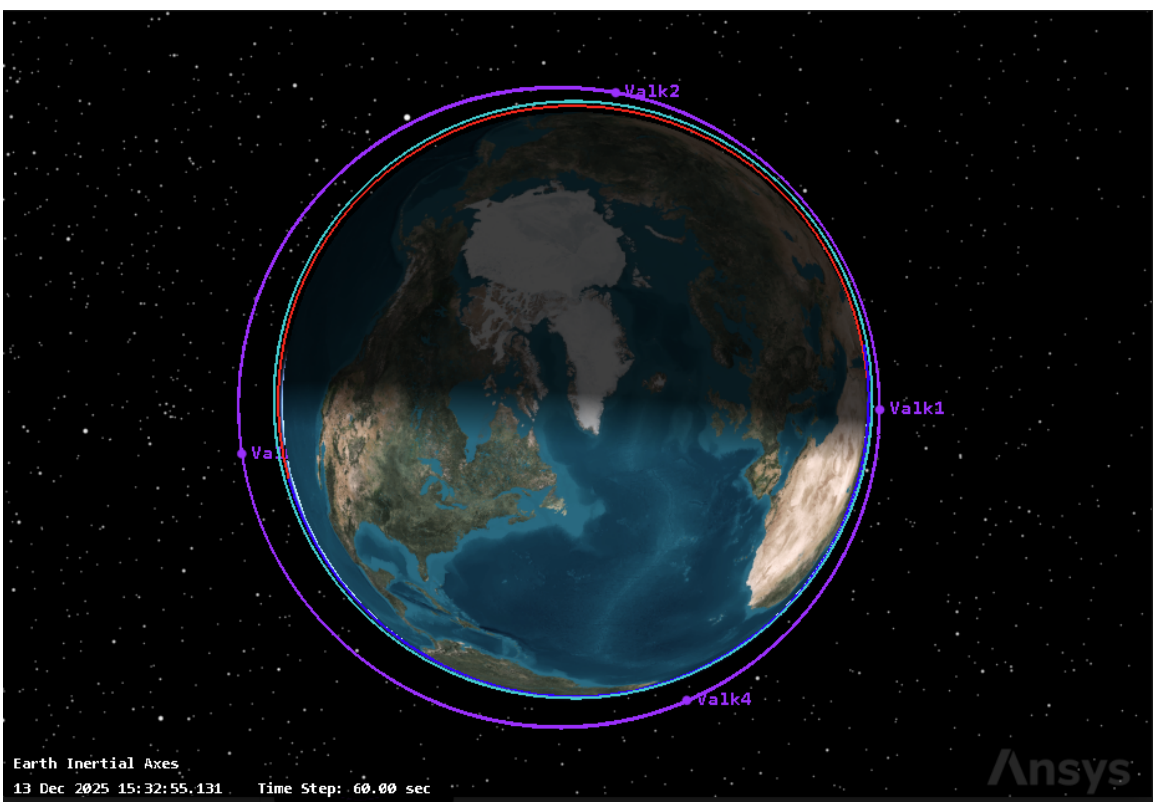

Figure 1: STK 3D visualization of Valkyrie-4: four satellites in a common 800 km orbit at $i = 28.5°$.

3 Background, research context, and modeling assumptions

3.1 Research context: where Valkyrie-4 fits

Earth-observation (EO) mission design typically trades off: (i) revisit time; (ii) resolution and SNR; (iii) coverage latitude band; and (iv) operations complexity (number of planes, cross-plane phasing, station-keeping). Rather than being a worldwide mapping system, Valkyrie-4 is positioned as a *fast-to-operate, fast-revisit* demonstrator focused on a low-latitude customer region of interest (**India AOI**) using a single orbital plane.

In order to shorten revisit times globally, many EO architectures employ large numbers of spacecraft and/or multiple orbital planes. Valkyrie-4 intentionally restricts itself to one plane to highlight the course-relevant orbital mechanics concepts: staged deployment, in-plane transfers, true-anomaly (along-track) phasing, and Astrogator targeting with transparent ΔV accounting.

3.2 Existing EO constellations and what they optimize for

Modern EO systems demonstrate that "revisit" can be achieved through very different architectural choices depending on the product being delivered. For example, PlanetScope achieves near-daily global land imaging using a large fleet (on the order of ~100+ spacecraft) and standardized tasking/processing; their published specifications indicate a reference sun-synchronous option near 475 km altitude with an inclination near 98° for broad latitude coverage. By contrast, Copernicus Sentinel-2 uses a smaller constellation (two spacecraft in the operational system) in a higher sun-synchronous orbit near 786 km and $i \approx 98.62°$, prioritizing consistent lighting geometry and systematic global coverage. Likewise, Landsat missions operate in sun-synchronous LEO (Landsat 8 is near 705 km altitude and $i \approx 98.2°$), emphasizing long-term continuity and repeatable observations rather than rapid intra-day access.

Valkyrie-4 differs intentionally: instead of sun-synchrony and global uniformity, it targets **time-distributed access over India** using a **low-inclination single-plane** architecture. The central design lever is not RAAN control or multi-plane geometry, but *in-plane timing*: loitering in a deployment orbit and inserting at different epochs to create along-track separation.

3.3 Why a low-inclination LEO plane for India AOI?

A low-inclination LEO plane concentrates pass density into a latitude belt that repeatedly intersects the India AOI. In contrast to Sun-synchronous designs, the mission objective here is to distribute access *in time* over the customer AOI (and the surrounding low-latitude belt), rather than to match a prescribed J_2-driven nodal precession rate.

3.4 Perturbations and operational implications

In LEO, atmospheric drag and the J_2 perturbation dominate long-term evolution:

- **Drag:** decreases semi-major axis and changes period; if spacecraft have different ballistic coefficients, differential drag can gradually erode along-track phasing.
- **J_2:** drives nodal regression and argument-of-perigee rotation. In a single-plane constellation, common-mode nodal regression is less concerning than *relative* drift; however, operational control is still motivated by maintaining consistent periods across the fleet.

3.5 Station maintenance and end-of-life disposal (mission realism)

A realistic operations concept budgets propellant for:

- **Orbit maintenance:** small maneuvers to maintain the operational altitude (and therefore keep periods aligned to preserve phasing).
- **Collision avoidance margin:** contingency ΔV for conjunction screening responses.
- **End-of-life disposal:** a controlled orbit-lowering plan or planned deorbit sequence, depending on mission policy and spacecraft capability.

In this project, a notional *station-keeping + disposal reserve* is tracked separately from insertion propellant (Table 4). The modeling premise is that long-term drift is minimized by maintaining identical periods; the primary phasing is created during deployment by loiter timing.

4 Mission design and required tables

4.1 Constants and baseline orbits

Unless otherwise noted,

$$\mu_\oplus = 398600\ \mathrm{km}^3/\mathrm{s}^2, \qquad R_\oplus = 6378\ \mathrm{km}.$$

Baseline circular orbits are:

- Parking: $h = 200\,\mathrm{km}$, $i = 28.5°$
- Deployment: $h = 300\,\mathrm{km}$, $i = 28.5°$
- Final operational: $h = 800\,\mathrm{km}$, $i = 28.5°$

Baseline circular orbits (analytic). Periods computed from $T = 2\pi\sqrt{r^3/\mu_\oplus}$.

Phase	h (km)	$r = R_\oplus + h$ (km)	i (deg)	T (min)
Parking	200	6578	28.5	88.49
Deployment	300	6678	28.5	90.52
Operational	800	7178	28.5	100.87

4.2 Astrogator implementation overview (what was actually built)

Upper stage sequence: A parking-orbit starting state, an impulsive maneuver to boost apogee toward the deployment altitude, a coast segment to apogee, and a second impulsive maneuver to circularize at 300 km were all included in the creation of an Astrogator MCS. The final circular orbit (altitude and eccentricity) is converged using a differential corrector/targeting loop.

Satellite sequences: Each satellite begins in the deployment orbit after separation, executes a *Repeat Count* loiter (an integer number of revolutions), and then performs a finite maneuver that targets the final 800 km circular orbit. Finite burn duration and propellant usage are taken from STK outputs and summarized in Table 4 and the Appendix MCS text printouts.

4.3 Required Table 1: Mission sequenced events timeline

Mission elapsed time (MET) is measured from the start of upper-stage Burn 1. Deployment orbit period for loiter timing uses $T_{\text{deploy}} \approx 5431$ s (analytic).

Table 1: Mission sequenced events timeline (MET from upper-stage Burn 1).

Description	Duration (s)	MET (s)
Upper stage: Burn 1 (raise apogee)	0	0
Upper stage: coast to apogee (transfer)	2685	2685
Upper stage: Burn 2 (circularize at 300 km)	0	2685
Deploy Valk1–Valk4 into 300 km orbit	0	2685
Valk1: loiter (1 rev)	5431	8116
Valk1: finite insertion to 800 km	764	8880
Valk2: loiter (3 rev)	16293	18978
Valk2: finite insertion to 800 km	766	19744
Valk3: loiter (6 rev)	32586	35271
Valk3: finite insertion to 800 km	764	36035
Valk4: loiter (9 rev)	48879	51564
Valk4: finite insertion to 800 km	763	52327

4.4 Required Table 2: Orbital data by phase

Table 2: Mission orbital data for each key phase (STK-reported apoapsis/periapsis altitudes; osculating values).

Phase	Object	Apo (km)	Peri (km)	e	T (s)
Initial park orbit	Upper stage	200.14	200.14	0.0000	5310
Deployment orbit	Upper stage	301.14	293.08	0.0006	5427
Deployment orbit	Valk1	301.14	293.08	0.0006	5427
Deployment orbit	Valk2	301.14	293.08	0.0006	5427
Deployment orbit	Valk3	301.14	293.08	0.0006	5427
Deployment orbit	Valk4	301.14	293.08	0.0006	5427
Final orbit	Valk1	1278.86	311.93	0.0674	6046
Final orbit	Valk2	1278.86	311.93	0.0674	6046
Final orbit	Valk3	1278.86	311.93	0.0674	6046
Final orbit	Valk4	1278.86	311.93	0.0674	6046

Note: The "Final orbit" entries reflect the single finite-burn insertion result reported by STK (osculating elements), so the orbit is slightly non-circular rather than an exactly circular 800 km target.

4.5 Required Table 3: Mission ΔV and phasing requirements

Table 3: Mission design ΔV and phasing requirements (STK totals for burns).

Object	ΔV_{total} (m/s)	Burn model / # burns	N_{loiter} (rev)
Upper stage	60.534	Impulsive / 2	—
Valk1	267.017	Finite / 1	1
Valk2	266.580	Finite / 1	3
Valk3	266.204	Finite / 1	6
Valk4	265.969	Finite / 1	9

4.6 Required Table 4: Satellite design details

Table 4: Satellite design details used in STK (and inferred effective performance).

Parameter	Value	Units
Dry mass (STK)	500	kg
Initial fuel mass (STK)	1000	kg
Initial total mass (STK)	1500	kg
Typical insertion fuel used (finite burn)	129	kg
Typical insertion duration (finite burn)	764	s
Inferred I_{sp} (from ΔV and fuel used)	301	s
Inferred thrust (from $\dot{m}$ and inferred I_{sp})	510	N
Notional station-keeping + disposal reserve (concept)	50	kg

4.7 Analytic checks (impulsive sanity checks)

Circular speed at radius r is

$$v_c(r) = \sqrt{\frac{\mu_\oplus}{r}}, \qquad r = R_\oplus + h.$$

The impulsive Hohmann estimate for 200 km→300 km gives $\Delta V \approx 0.060\,\text{km/s}$, consistent with STK 60.534 m/s. For 300 km→800 km, the first-cut impulsive estimate is $\Delta V \approx 0.27\,\text{km/s}$ per satellite, consistent with the finite-burn totals.

4.8 Phasing strategy via loiter (true-anomaly spacing by timing)

Phasing is generated by controlling *when* each satellite inserts into the final orbit. Valkyrie-4 uses the 300 km orbit as a "waiting room" with loiter revolution counts $\{1, 3, 6, 9\}$. After insertion, all satellites share the same final orbit; the primary separation is along-track (true anomaly / mean anomaly).

The most equally spaced design for a four-satellite single-plane constellation is roughly $360°/4 = 90°$ spacing. In actuality, the accomplished separation in STK is confirmed visually in the 3D view and by examining relative true anomaly/along-track angles at a shared epoch.

5 Results (STK outputs, evidence, and verification)

This section includes STK visual outputs and MCS evidence exported to the `figs/` folder.

5.1 India AOI ground-track evidence

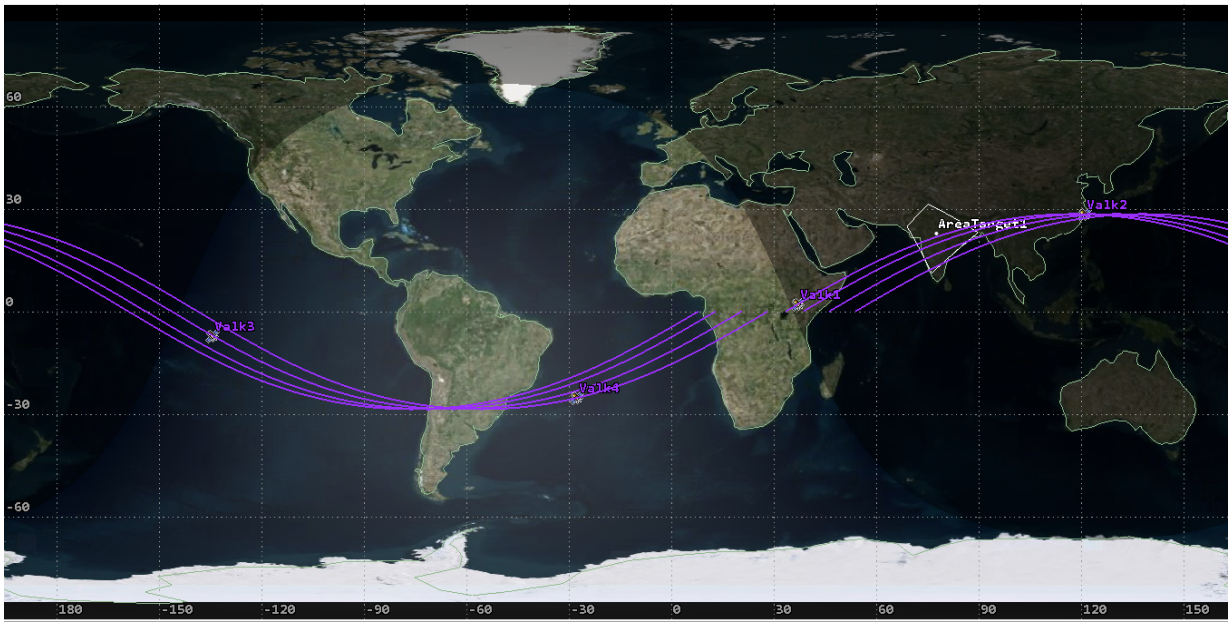

Figure 2: STK ground-track view illustrating repeated passes that intersect the **India AOI**. The displayed arcs show along-track separation between phased spacecraft (e.g., Valk1 vs. Valk2) over the same AOI, demonstrating time-distributed access without any plane changes.

5.2 Upper stage Astrogator MCS evidence

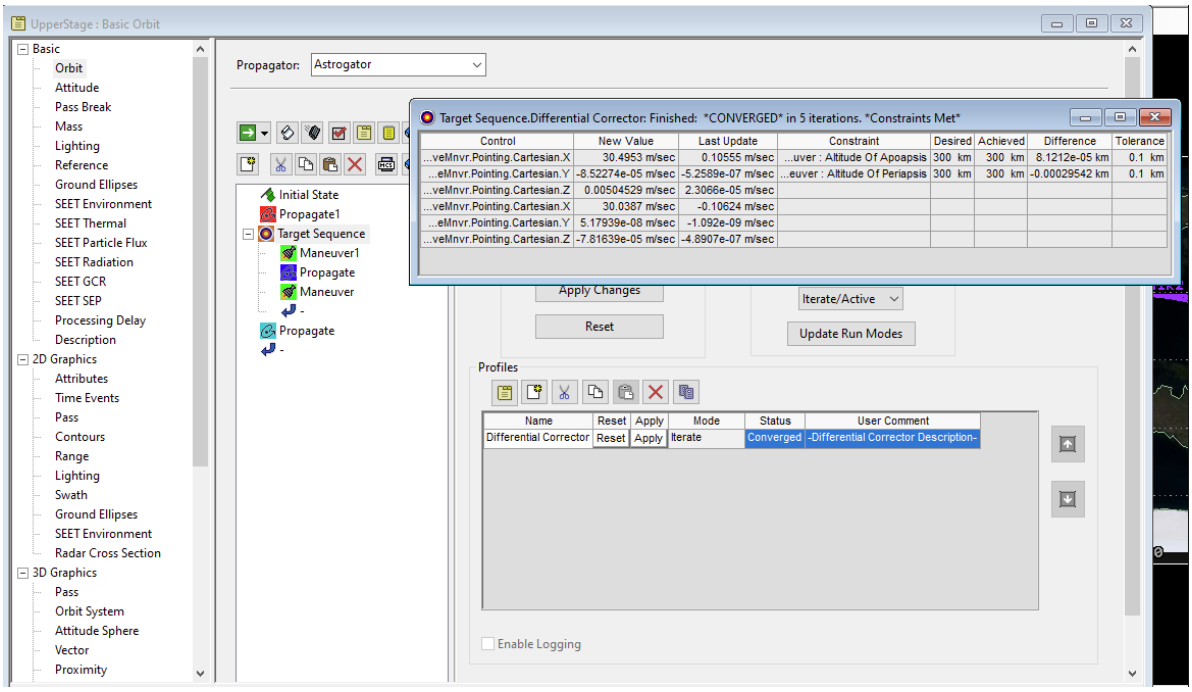

Figure 3: Upper stage Astrogator MCS output screenshot (exported from STK), showing the two-burn delivery to the 300 km deployment orbit and associated ΔV.

5.3 Satellite Astrogator MCS evidence (Valk1 shown)

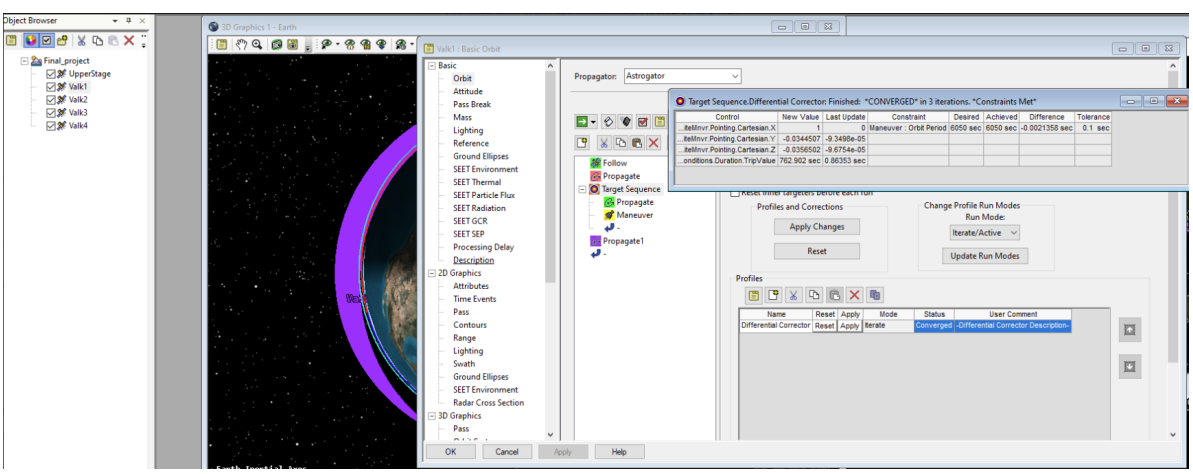

Figure 4: Valk1 Astrogator MCS output screenshot (exported from STK), showing loiter logic and the finite insertion burn to the 800 km operational orbit.

5.4 Operational orbit period check

Using

$$T = 2\pi\sqrt{\frac{r^3}{\mu_\oplus}}, \qquad r = R_\oplus + 800\,\text{km},$$

gives $T \approx 6052\,\text{s}$, consistent with the reported $\sim 6050\,\text{s}$.

6 Discussion and lessons learned

6.1 What worked well

- **Phasing by timing is efficient:**Along-track separation was accomplished without plane adjustments by loitering in the deployment orbit.
- **Astrogator workflow is transparent:** MCS segment summaries exhibit targeting logic, repeat-count behavior, and clearly attribute ΔV.
- **Separation of responsibilities:** Upper-stage delivery vs. satellite insertion provides a clean architecture for mission cost accounting.

6.2 Limitations and what would be improved next

- **Long-term drift not modeled:**Phasing can be undermined by differential drag/BC discrepancies; a further study would incorporate station-keeping maneuvers and simulate disturbances.
- **Scalability path:**In a growth version, a second plane or more satellites would be added to each plane, and the revisit improvement would be assessed for each additional spacecraft.

7 Conclusions

In STK Astrogator, Valkyrie-4 shows a full deployment strategy: parking 200 km → deployment 300 km (higher stage) → satellites with operational 800 km. The main concept of mission design is that, instead of costly plane-change maneuvers, constellation spacing can be achieved using *timing* (loiter). With the use of STK screenshots and accompanying MCS text summaries, Astrogator targeting and repeat-count logic offer a clear approach for converging burns and recording total ΔV.

8 References

References

[1] Planet Labs PBC. "PlanetScope" (product/mission overview and documentation). `https://www.planet.com/`.

[2] European Space Agency (ESA). "Sentinel-2 Mission" (Copernicus factsheets / mission description). `https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-2`.

[3] U.S. Geological Survey (USGS). "Landsat Missions" (program and mission overview). https://www.usgs.gov/landsat-missions.

[4] Vallado, D. A. *Fundamentals of Astrodynamics and Applications*. Microcosm Press, latest edition.

Appendix: STK Mission Control Sequence (MCS) Printouts

Upper Stage: Formatted STK MCS Summary

```
                                                                                                                    13 Dec 2025 01:44:13
Satellite-UpperStage, Valk1, Valk2, Valk3, Valk4


                    UpperStage - Astrogator Mission Control Sequence Summary
-----------------------------------------------------------------------------------------------------

***-------------------------------------------------------------------------------***
MCS Segment Type: InitialState
Name: Initial State
User Comment:  Initializes state data

----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 19:00:00.000  UTC Julian Date: 2461021.29166667
Julian Ephemeris Date: 2461021.29246741
Time past epoch: 0 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6578.1369900981526371 km             Vx:        0.0000000000000000 km/sec
         Y:        0.0000000000000000 km             Vy:        6.8409424392281473 km/sec
         Z:        0.0000000000000000 km             Vz:        3.7143286882773934 km/sec

Parameter Set Type:  Keplerian
       sma:     6578.1369950490825431 km           RAAN:                        0 deg
       ecc:        0.0000000007526338                 w:                        0 deg
       inc:                     28.5 deg            TA:                        0 deg

Parameter Set Type:  Spherical
 Right Asc:                         0 deg     Horiz. FPA:                        0 deg
      Decl:                         0 deg        Azimuth:                     61.5 deg
       |R|:     6578.1369900981526371 km            |V|:        7.7842617544243211 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:                         0 deg       Mean Anom:                        0 deg
 Long Peri:                         0 deg        Arg. Lat:                        0 deg
 True Long:                         0 deg        Vert FPA:                       90 deg
  Ang. Mom:         51205.94018738497 km^2/sec         p:     6578.1369950490825431 km
        C3:        -60.59473097018192 km^2/sec^2   Energy:       -30.29736548509096 km^2/sec^2
   Vel. RA:                        90 deg       Vel. Decl:                     28.5 deg
 Rad. Peri:     6578.1369900981544561 km        Vel. Peri:        7.7842617544243193 km/sec
  Rad. Apo:     6578.1370000000106302 km         Vel. Apo:         7.784261742706923 km/sec
 Mean Mot.:       0.06780116393951884 deg/sec
    Period:         5309.643361301783 sec          Period:         88.49405602169638 min
    Period:          1.47490093369494 hr           Period:       0.06145420557062248 day
               Time Past Periapsis:                                 0 sec
          Time Past Ascending Node:          -5309.643361301783 sec
   Beta Angle (Orbit plane to Sun):            4.98118336335869 deg
Mean Sidereal Greenwich Hour Angle:            5.74256784219948 deg

Geodetic Parameters:
  Latitude:          0.14588789134264 deg
 Longitude:        -5.410135141421479 deg
  Altitude:      200.0001276094552622 km
Geocentric Parameters:
  Latitude:        0.1449409596110963 deg
 Longitude:        -5.410135141421479 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate1
User Comment:  Propagates until stopping conditions are met
```

```
Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  11 Dec 2025 19:46:38.383 [2461021.32405536]:  Stopped on: Periapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    45
  Average step size:  63.6255 sec
  Largest step size:  65.607 sec
  Smallest step size:  37.61 sec

----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 19:46:38.383  UTC Julian Date: 2461021.32405536
Julian Ephemeris Date: 2461021.3248561
Time past epoch: 2798.38 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6454.2370773189504689 km              Vx:        1.3907737021279203 km/sec
         Y:    -1020.9283229690486223 km              Vy:       -6.7511413062808163 km/sec
         Z:     -569.0316792017935086 km              Vz:       -3.6622773741696073 km/sec

Parameter Set Type:  Keplerian
       sma:     6575.9720431658824964 km            RAAN:         359.7591469923449 deg
       ecc:        0.0025486110173133                  w:         190.475488458513 deg
       inc:         28.49943199156019 deg             TA:     9.438760011667595e-06 deg

Parameter Set Type:  Spherical
 Right Asc:         188.9885476252715 deg     Horiz. FPA:     2.399457523905363e-08 deg
      Decl:        -4.976839625286338 deg       Azimuth:         118.0974686678057 deg
       |R|:     6559.2124483671259441 km            |V|:        7.8054106870334694 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:     9.430238149809192e-06 deg       Mean Anom:     9.406204140964701e-06 deg
 Long Peri:          190.234635450858 deg        Arg. Lat:         190.475497897273 deg
 True Long:          190.234644889618 deg        Vert FPA:         89.99999997600543 deg
  Ang. Mom:         51197.34694300772 km^2/sec          p:     6575.9293294779336065 km
        C3:         -60.6146800630407 km^2/sec^2   Energy:       -30.30734003152035 km^2/sec^2
   Vel. RA:         281.6404146250196 deg       Vel. Decl:        -27.9822021087112 deg
 Rad. Peri:     6559.2124483671250346 km        Vel. Peri:        7.8054106870334694 km/sec
  Rad. Apo:     6592.7316379646381392 km         Vel. Apo:        7.765725916733021 km/sec
 Mean Mot.:       0.06783464910427357 deg/sec
    Period:         5307.022366204294 sec           Period:        88.45037277007157 min
    Period:         1.474172879501193 hr            Period:       0.06142386997921637 day
              Time Past Periapsis:       0.0001386637104366198 sec
         Time Past Ascending Node:           2808.722375016442 sec
   Beta Angle (Orbit plane to Sun):            5.0016356935073 deg
Mean Sidereal Greenwich Hour Angle:           17.4344212199642 deg

Geodetic Parameters:
  Latitude:        -5.153663216624388 deg
 Longitude:         171.8883991446942 deg
  Altitude:      181.2466003139180089 km
Geocentric Parameters:
  Latitude:        -5.120294825926745 deg
 Longitude:         171.8883991446942 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: TargeterSequence
Name: Target Sequence
User Comment:  Sequence that runs targeting profiles

Sequence Start: 11 Dec 2025 19:46:38.383 UTCG;   2461021.32405536 UTC Julian Date
Sequence Stop:  11 Dec 2025 20:31:03.079 UTCG;   2461021.35489675 UTC Julian Date

***<<<< Start of Sequence: Target Sequence >>>>***
```

```
***-------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Impulsive
Name: Target Sequence.Maneuver1
User Comment:  Maneuvers satellite with an impulsive burn or finite burn

Maneuver Summary:
  Maneuver Start: 11 Dec 2025 19:46:38.383 UTCG;   2461021.32405536 UTC Julian Date
  Maneuver Stop:  11 Dec 2025 19:46:38.383 UTCG;   2461021.32405536 UTC Julian Date
  Duration:  0 sec
  Fuel Used:                      0 kg
  DeltaV Magnitude:        30.49528318334543 m/sec
  Estimated Equivalent Finite Burn Duration:        60.67555650860218 sec
  Estimated Fuel Used:  10.312 kg  (Update mass OFF)
  Maneuver Direction Specification:  Thrust Vector

DeltaV vector with respect to VNC(Earth) axes:

       X (Velocity):         30.49528276586725 m/sec
         Y (Normal):    -8.522742094818625e-05 m/sec
      Z (Co-Normal):      0.005045291668791529 m/sec
           Azimuth:    -0.0001601287502921223 deg
         Elevation:      0.009479299425260015 deg
         Magnitude:         30.49528318334545 m/sec

DeltaV vector with respect to Earth Inertial axes:

              X:         5.428707088792987 m/sec
              Y:        -26.37705861690212 m/sec
              Z:         -14.3088159711777 m/sec
        Azimuth:        -78.37024994635073 deg
      Elevation:        -27.98329227609543 deg
      Magnitude:         30.49528318334544 m/sec

DeltaV vector with respect to spacecraft body axes:

              X:         30.49528318334544 m/sec
              Y:                         0 m/sec
              Z:                         0 m/sec
        Azimuth:                         0 deg
      Elevation:                         0 deg
      Magnitude:         30.49528318334544 m/sec

Attitude with respect to Earth Inertial axes:
  ---  -----------------------
  qx:      -0.3925249060985801
  qy:       0.5508920234536523
  qz:      -0.3278856738352152
  qs:       0.6594946258166692


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Satellite State at Beginning of Segment:
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UTC Gregorian Date: 11 Dec 2025 19:46:38.383  UTC Julian Date: 2461021.32405536
Julian Ephemeris Date: 2461021.3248561
Time past epoch: 2798.38 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6454.2370773189504689 km              Vx:        1.3907737021279203 km/sec
         Y:    -1020.9283229690486223 km              Vy:       -6.7511413062808163 km/sec
         Z:     -569.0316792017935086 km              Vz:       -3.6622773741696073 km/sec

Parameter Set Type:  Keplerian
       sma:     6575.9720431658824964 km            RAAN:         359.7591469923449 deg
       ecc:        0.0025486110173133                  w:         190.475488458513 deg
       inc:         28.49943199156019 deg             TA:     9.438760011667595e-06 deg

Parameter Set Type:  Spherical
 Right Asc:         188.9885476252715 deg     Horiz. FPA:     2.399457523905363e-08 deg
      Decl:        -4.976839625286338 deg        Azimuth:         118.0974686678057 deg
       |R|:     6559.2124483671259441 km             |V|:        7.8054106870334694 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:     9.430238149809192e-06 deg       Mean Anom:     9.406204140964701e-06 deg
 Long Peri:          190.234635450858 deg        Arg. Lat:          190.475497897273 deg
 True Long:          190.234644889618 deg        Vert FPA:         89.99999997600543 deg
  Ang. Mom:         51197.34694300772 km^2/sec          p:     6575.9293294779336065 km
        C3:         -60.6146800630407 km^2/sec^2   Energy:        -30.30734003152035 km^2/sec^2
   Vel. RA:         281.6404146250196 deg       Vel. Decl:         -27.9822021087112 deg
 Rad. Peri:     6559.2124483671250346 km        Vel. Peri:        7.8054106870334694 km/sec
  Rad. Apo:     6592.7316379646381392 km         Vel. Apo:         7.765725916733021 km/sec
 Mean Mot.:       0.06783464910427357 deg/sec
    Period:         5307.022366204294 sec          Period:         88.45037277007157 min
    Period:         1.474172879501193 hr           Period:       0.06142386997921637 day
              Time Past Periapsis:       0.0001386637104366198 sec
          Time Past Ascending Node:           2808.722375016442 sec
   Beta Angle (Orbit plane to Sun):             5.0016356935073 deg
Mean Sidereal Greenwich Hour Angle:            17.4344212199642 deg
```

```
Geodetic Parameters:
  Latitude:        -5.153663216624388 deg
 Longitude:         171.8883991446942 deg
  Altitude:      181.2466003139180089 km
Geocentric Parameters:
  Latitude:        -5.120294825926745 deg
 Longitude:         171.8883991446942 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 19:46:38.383  UTC Julian Date: 2461021.32405536
Julian Ephemeris Date: 2461021.3248561
Time past epoch: 2798.38 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6454.2370773189504689 km              Vx:        1.3962024092167136 km/sec
         Y:    -1020.9283229690486223 km              Vy:       -6.7775183648977180 km/sec
         Z:     -569.0316792017935086 km              Vz:       -3.6765861901407852 km/sec

Parameter Set Type:  Keplerian
       sma:     6628.1298309028470612 km            RAAN:         359.7591472298033 deg
       ecc:        0.0103977116360476                  w:         190.4719104797463 deg
       inc:         28.49943260435244 deg             TA:      0.003587208843099013 deg

Parameter Set Type:  Spherical
 Right Asc:         188.9885476252715 deg     Horiz. FPA:      3.69149322801768e-05 deg
      Decl:        -4.976839625286338 deg        Azimuth:         118.0974692909846 deg
       |R|:     6559.2124483671259441 km            |V|:        7.8359059698009599 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:      0.003550102007634313 deg       Mean Anom:      0.003513189070703997 deg
 Long Peri:         190.2310577095496 deg        Arg. Lat:         190.4754976885894 deg
 True Long:         190.2346449183927 deg        Vert FPA:         89.99996308506772 deg
  Ang. Mom:         51397.37198134207 km^2/sec          p:     6627.4132478311539671 km
        C3:        -60.13769368873464 km^2/sec^2   Energy:        -30.06884684436732 km^2/sec^2
   Vel. RA:         281.6403731217335 deg       Vel. Decl:        -27.98220635294986 deg
 Rad. Peri:     6559.2124482348344827 km        Vel. Peri:        7.8359059699573752 km/sec
  Rad. Apo:     6697.0472135708614587 km         Vel. Apo:         7.674631870175666 km/sec
 Mean Mot.:       0.06703552423318408 deg/sec
    Period:         5370.286935442387 sec          Period:         89.50478225737312 min
    Period:         1.491746370956219 hr           Period:       0.06215609878984244 day
              Time Past Periapsis:         0.05240787046705738 sec
          Time Past Ascending Node:          2844.665648778976 sec
   Beta Angle (Orbit plane to Sun):           5.00163627670941 deg
Mean Sidereal Greenwich Hour Angle:           17.4344212199642 deg

Geodetic Parameters:
  Latitude:        -5.153663216624388 deg
 Longitude:         171.8883991446942 deg
  Altitude:      181.2466003139180089 km
Geocentric Parameters:
  Latitude:        -5.120294825926745 deg
 Longitude:         171.8883991446942 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

```
***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Target Sequence.Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  11 Dec 2025 20:31:03.079 [2461021.35489675]:  Stopped on: Apoapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    42
  Average step size:  63.8074 sec
  Largest step size:  67.104 sec
  Smallest step size:  32.604 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 20:31:03.079  UTC Julian Date: 2461021.35489675
Julian Ephemeris Date: 2461021.35569749
Time past epoch: 5463.08 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6588.8495122584326964 km             Vx:       -1.2538463126693009 km/sec
         Y:      943.6679775811198851 km             Vy:        6.6752779344096886 km/sec
         Z:      542.2759587387773763 km             Vz:        3.6183791783841301 km/sec

Parameter Set Type:  Keplerian
       sma:     6626.7062376391704674 km           RAAN:         359.5203185157674 deg
       ecc:        0.0077610963637475                 w:         189.7987077438067 deg
       inc:         28.49884107538112 deg            TA:         179.9998003821099 deg

Parameter Set Type:  Spherical
 Right Asc:         8.150586049268426 deg     Horiz. FPA:     1.561371632842401e-06 deg
      Decl:         4.657641982846678 deg        Azimuth:         61.85279856822678 deg
       |R|:     6678.1367433234163400 km            |V|:        7.6957217956456230 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         179.9997988244043 deg       Mean Anom:         179.9997972630611 deg
 Long Peri:         189.3190262595742 deg        Arg. Lat:         9.798508125916689 deg
 True Long:         9.318826641684103 deg        Vert FPA:         89.99999843862837 deg
  Ang. Mom:         51393.08248989588 km^2/sec          p:     6626.3070805285160532 km
        C3:        -60.15061286947969 km^2/sec^2   Energy:        -30.07530643473984 km^2/sec^2
   Vel. RA:         100.6381589031147 deg       Vel. Decl:         28.04602104345153 deg
 Rad. Peri:     6575.2757319546053623 km        Vel. Peri:        7.8161106218154703 km/sec
  Rad. Apo:     6678.1367433237337536 km         Vel. Apo:         7.695721795645255 km/sec
 Mean Mot.:       0.06705712692140155 deg/sec
    Period:         5368.556878703739 sec          Period:         89.47594797839565 min
    Period:         1.491265799639927 hr           Period:       0.06213607498499698 day
              Time Past Periapsis:          2684.275416005237 sec
         Time Past Ascending Node:          148.3919031661899 sec
   Beta Angle (Orbit plane to Sun):          5.02131017527567 deg
Mean Sidereal Greenwich Hour Angle:          28.5677214128541 deg

Geodetic Parameters:
  Latitude:         4.832135620046327 deg
 Longitude:        -20.08315201296385 deg
  Altitude:      300.1502710525391535 km
Geocentric Parameters:
  Latitude:         4.801386190700405 deg
 Longitude:        -20.08315201296385 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Impulsive
Name: Target Sequence.Maneuver
User Comment:  Maneuvers satellite with an impulsive burn or finite burn
```

```
Maneuver Summary:
  Maneuver Start: 11 Dec 2025 20:31:03.079 UTCG;   2461021.35489675 UTC Julian Date
  Maneuver Stop:  11 Dec 2025 20:31:03.079 UTCG;   2461021.35489675 UTC Julian Date
  Duration:  0 sec
  Fuel Used:                        0 kg
  DeltaV Magnitude:        30.03866851251457 m/sec
  Estimated Equivalent Finite Burn Duration:        59.77167422281383 sec
  Estimated Fuel Used:  10.1584 kg  (Update mass OFF)
  Maneuver Direction Specification:  Thrust Vector

DeltaV vector with respect to VNC(Earth) axes:

       X (Velocity):         30.03866851241288 m/sec
         Y (Normal):      5.17939078292784e-08 m/sec
      Z (Co-Normal):    -7.816389557269154e-05 m/sec
            Azimuth:     9.879173978303843e-08 deg
          Elevation:    -0.0001490898747644485 deg
          Magnitude:         30.03866851251458 m/sec

DeltaV vector with respect to Earth Inertial axes:

              X:         -4.89420852762796 m/sec
              Y:         26.05556453757767 m/sec
              Z:         14.12359322272322 m/sec
        Azimuth:         100.6383271238087 deg
      Elevation:         28.04600742457278 deg
      Magnitude:         30.03866851251457 m/sec

DeltaV vector with respect to spacecraft body axes:

              X:         30.03866851251457 m/sec
              Y:                         0 m/sec
              Z:                         0 m/sec
        Azimuth:                         0 deg
      Elevation:                         0 deg
      Magnitude:         30.03866851251457 m/sec

Attitude with respect to Earth Inertial axes:
  ---  -----------------------
  qx:       0.5558717986170382
  qy:       0.3901082873237831
  qz:       0.6551951325867273
  qs:       0.3309704003354748


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Satellite State at Beginning of Segment:
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UTC Gregorian Date: 11 Dec 2025 20:31:03.079  UTC Julian Date: 2461021.35489675
Julian Ephemeris Date: 2461021.35569749
Time past epoch: 5463.08 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6588.8495122584326964 km              Vx:       -1.2538463126693009 km/sec
         Y:      943.6679775811198851 km              Vy:        6.6752779344096886 km/sec
         Z:      542.2759587387773763 km              Vz:        3.6183791783841301 km/sec

Parameter Set Type:  Keplerian
       sma:     6626.7062376391704674 km            RAAN:         359.5203185157674 deg
       ecc:        0.0077610963637475                  w:         189.7987077438067 deg
       inc:         28.49884107538112 deg             TA:         179.9998003821099 deg

Parameter Set Type:  Spherical
 Right Asc:         8.150586049268426 deg     Horiz. FPA:     1.561371632842401e-06 deg
      Decl:         4.657641982846678 deg        Azimuth:         61.85279856822678 deg
       |R|:     6678.1367433234163400 km            |V|:        7.6957217956456230 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         179.9997988244043 deg       Mean Anom:         179.9997972630611 deg
 Long Peri:         189.3190262595742 deg        Arg. Lat:         9.798508125916689 deg
 True Long:         9.318826641684103 deg        Vert FPA:         89.99999843862837 deg
  Ang. Mom:         51393.08248989588 km^2/sec          p:     6626.3070805285160532 km
        C3:        -60.15061286947969 km^2/sec^2   Energy:        -30.07530643473984 km^2/sec^2
   Vel. RA:         100.6381589031147 deg       Vel. Decl:         28.04602104345153 deg
 Rad. Peri:     6575.2757319546053623 km        Vel. Peri:        7.8161106218154703 km/sec
  Rad. Apo:     6678.1367433237337536 km         Vel. Apo:         7.695721795645255 km/sec
 Mean Mot.:       0.06705712692140155 deg/sec
    Period:         5368.556878703739 sec          Period:         89.47594797839565 min
    Period:         1.491265799639927 hr           Period:        0.06213607498499698 day
              Time Past Periapsis:          2684.275416005237 sec
          Time Past Ascending Node:         148.3919031661899 sec
   Beta Angle (Orbit plane to Sun):          5.02131017527567 deg
Mean Sidereal Greenwich Hour Angle:           28.5677214128541 deg

Geodetic Parameters:
  Latitude:         4.832135620046327 deg
 Longitude:        -20.08315201296385 deg
  Altitude:      300.1502710525391535 km
```

```
Geocentric Parameters:
  Latitude:         4.801386190700405 deg
 Longitude:        -20.08315201296385 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Altitude Of Apoapsis =      299.9997433237340374 km
    Altitude Of Periapsis =      197.1387319546053050 km

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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 20:31:03.079  UTC Julian Date: 2461021.35489675
Julian Ephemeris Date: 2461021.35569749
Time past epoch: 5463.08 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6588.8495122584326964 km             Vx:       -1.2587405211969289 km/sec
         Y:      943.6679775811198851 km             Vy:        6.7013334989472666 km/sec
         Z:      542.2759587387773763 km             Vz:        3.6325027716068536 km/sec

Parameter Set Type:  Keplerian
       sma:     6678.1368928938627505 km           RAAN:         359.5203185159044 deg
       ecc:        0.0000000281991678                 w:         332.3824562085724 deg
       inc:         28.49884107575962 deg            TA:         37.41605191722394 deg

Parameter Set Type:  Spherical
 Right Asc:         8.150586049268426 deg     Horiz. FPA:     9.816926039853122e-07 deg
      Decl:         4.657641982846678 deg        Azimuth:         61.85279856784268 deg
       |R|:     6678.1367433234163400 km            |V|:        7.7257604641580375 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         37.41605093553132 deg       Mean Anom:         37.41604995383871 deg
 Long Peri:         331.9027747244768 deg        Arg. Lat:         9.798508125796285 deg
 True Long:         9.318826641700692 deg        Vert FPA:         89.99999901830739 deg
  Ang. Mom:         51593.68482580916 km^2/sec          p:     6678.1368928938591125 km
        C3:        -59.68737207590736 km^2/sec^2   Energy:        -29.84368603795368 km^2/sec^2
   Vel. RA:         100.6381595571767 deg       Vel. Decl:         28.04602099050011 deg
 Rad. Peri:     6678.1367045759607208 km        Vel. Peri:        7.7257605089839476 km/sec
  Rad. Apo:     6678.1370812117665992 km         Vel. Apo:         7.725760073263927 km/sec
 Mean Mot.:       0.06628397490356727 deg/sec
    Period:         5431.177000530569 sec          Period:         90.51961667550948 min
    Period:         1.508660277925158 hr           Period:       0.06286084491354825 day
             Time Past Periapsis:            564.481083222199 sec
         Time Past Ascending Node:          147.8261929015524 sec
   Beta Angle (Orbit plane to Sun):           5.02131017563728 deg
Mean Sidereal Greenwich Hour Angle:           28.5677214128541 deg

Geodetic Parameters:
  Latitude:         4.832135620046327 deg
 Longitude:        -20.08315201296385 deg
  Altitude:      300.1502710525391535 km
Geocentric Parameters:
  Latitude:         4.801386190700405 deg
 Longitude:        -20.08315201296385 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

```
User-selected results:
    Altitude Of Apoapsis =      300.0000812117662008 km
    Altitude Of Periapsis =      299.9997045759605498 km


***<<<< End of Sequence: Target Sequence >>>>***


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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 20:31:03.079  UTC Julian Date: 2461021.35489675
Julian Ephemeris Date: 2461021.35569749
Time past epoch: 5463.08 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6588.8495122584326964 km              Vx:       -1.2587405211969289 km/sec
         Y:      943.6679775811198851 km              Vy:        6.7013334989472666 km/sec
         Z:      542.2759587387773763 km              Vz:        3.6325027716068536 km/sec

Parameter Set Type:  Keplerian
       sma:     6678.1368928938627505 km            RAAN:         359.5203185159044 deg
       ecc:        0.0000000281991678                  w:         332.3824562085724 deg
       inc:         28.49884107575962 deg             TA:         37.41605191722394 deg

Parameter Set Type:  Spherical
 Right Asc:         8.150586049268426 deg     Horiz. FPA:     9.816926039853122e-07 deg
      Decl:         4.657641982846678 deg        Azimuth:         61.85279856784268 deg
       |R|:     6678.1367433234163400 km            |V|:        7.7257604641580375 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         37.41605093553132 deg       Mean Anom:         37.41604995383871 deg
 Long Peri:         331.9027747244768 deg        Arg. Lat:         9.798508125796285 deg
 True Long:         9.318826641700692 deg        Vert FPA:         89.99999901830739 deg
  Ang. Mom:         51593.68482580916 km^2/sec          p:     6678.1368928938591125 km
        C3:        -59.68737207590736 km^2/sec^2   Energy:        -29.84368603795368 km^2/sec^2
   Vel. RA:         100.6381595571767 deg       Vel. Decl:         28.04602099050011 deg
 Rad. Peri:     6678.1367045759607208 km        Vel. Peri:        7.7257605089839476 km/sec
  Rad. Apo:     6678.1370812117665992 km         Vel. Apo:         7.725760073263927 km/sec
 Mean Mot.:       0.06628397490356727 deg/sec
    Period:         5431.177000530569 sec          Period:         90.51961667550948 min
    Period:         1.508660277925158 hr           Period:       0.06286084491354825 day
             Time Past Periapsis:           564.481083222199 sec
         Time Past Ascending Node:          147.8261929015524 sec
   Beta Angle (Orbit plane to Sun):           5.02131017563728 deg
Mean Sidereal Greenwich Hour Angle:           28.5677214128541 deg

Geodetic Parameters:
  Latitude:         4.832135620046327 deg
 Longitude:        -20.08315201296385 deg
  Altitude:      300.1502710525391535 km
Geocentric Parameters:
  Latitude:         4.801386190700405 deg
 Longitude:        -20.08315201296385 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  11 Dec 2025 23:28:57.781 [2461021.47844654]:  Stopped on: AscendingNode; Run Sequence STOP

Propagation Statistics:
  Number of steps:    161
  Average step size:  66.3314 sec
  Largest step size:  68.938 sec
```

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  Smallest step size:  42.46 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
               Time Past Periapsis:           1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  500 kg
     Total Mass:  1000 kg
Area/Mass Ratio:  2e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


                              Valk1 - Astrogator Mission Control Sequence Summary
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***-------------------------------------------------------------------------------***
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

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Satellite State at Beginning of Segment:
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UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
```

```
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                 w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
             Time Past Periapsis:            1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                 w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
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```
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
             Time Past Periapsis:           1159.77625053902 sec
         Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 00:18:30.268 [2461021.51285032]:  Stopped on: Periapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    46
  Average step size:  65.5405 sec
  Largest step size:  67.093 sec
  Smallest step size:  41.067 sec

----------------------------------
Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 00:18:30.268  UTC Julian Date: 2461021.51285032
Julian Ephemeris Date: 2461021.51365106
Time past epoch: 19110.3 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6393.5128801469027167 km             Vx:        2.1695780132367246 km/sec
         Y:    -1598.2285579778906595 km             Vy:       -6.5494790827756679 km/sec
         Z:     -967.1466512763757919 km             Vz:       -3.5192805515574657 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.0869133897404026 km           RAAN:         358.3474406759541 deg
       ecc:        0.0024340813108811                 w:         197.7214414736106 deg
       inc:         28.49081375536728 deg            TA:         359.9999717332493 deg

Parameter Set Type:  Spherical
 Right Asc:         194.0349811514336 deg     Horiz. FPA:    -6.863650997038867e-08 deg
      Decl:        -8.348804446351229 deg        Azimuth:         117.3386709549386 deg
       |R|:     6660.8343409227300072 km            |V|:        7.7451972610003184 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.9999717996096 deg       Mean Anom:         359.9999718682516 deg
 Long Peri:         196.0688821495646 deg        Arg. Lat:         197.7214132068599 deg
 True Long:         196.0688538828139 deg        Vert FPA:         90.00000006863651 deg
  Ang. Mom:         51589.47589329159 km^2/sec          p:     6677.0473533068443430 km
        C3:        -59.69675798298745 km^2/sec^2   Energy:        -29.84837899149372 km^2/sec^2
   Vel. RA:         288.3279493428208 deg       Vel. Decl:        -27.02519741701506 deg
 Rad. Peri:     6660.8343409227300072 km        Vel. Peri:        7.7451972610003192 km/sec
  Rad. Apo:     6693.3394858567498886 km         Vel. Apo:         7.707583935089783 km/sec
 Mean Mot.:       0.06629961036375588 deg/sec
    Period:         5429.896164168135 sec          Period:         90.49826940280225 min
    Period:         1.508304490046704 hr           Period:       0.06284602041861267 day
             Time Past Periapsis:           5429.895739855712 sec
         Time Past Ascending Node:          2983.523764673164 sec
   Beta Angle (Orbit plane to Sun):           5.10479436255779 deg
Mean Sidereal Greenwich Hour Angle:           85.5866923411367 deg
```

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Geodetic Parameters:
  Latitude:        -8.543839813941487 deg
 Longitude:         108.7856649840746 deg
  Altitude:      283.1656131524050011 km
Geocentric Parameters:
  Latitude:        -8.489875882786237 deg
 Longitude:         108.7856649840746 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: TargeterSequence
Name: Target Sequence
User Comment:  Sequence that runs targeting profiles

Sequence Start: 12 Dec 2025 00:18:30.268 UTC;   2461021.51285032 UTC Julian Date
Sequence Stop:  12 Dec 2025 01:08:36.391 UTC;   2461021.54764341 UTC Julian Date

***<<<< Start of Sequence: Target Sequence >>>>***


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Target Sequence.Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 01:02:14.940 [2461021.54322847]:  Stopped on: Apoapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    40
  Average step size:  66.4113 sec
  Largest step size:  67.099 sec
  Smallest step size:  60 sec

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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 12 Dec 2025 01:02:14.940  UTC Julian Date: 2461021.54322847
Julian Ephemeris Date: 2461021.54402921
Time past epoch: 21734.9 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6560.1089303407779880 km              Vx:       -1.4343353683819131 km/sec
         Y:     1040.4569155812102963 km              Vy:        6.6842841697523925 km/sec
         Z:      681.8183770497626028 km              Vz:        3.6002034962275498 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.1200288823310984 km            RAAN:         358.1100331230998 deg
       ecc:        0.0000165510830409                  w:         12.47262363527748 deg
       inc:         28.49060592937986 deg             TA:         359.8883242669613 deg

Parameter Set Type:  Spherical
 Right Asc:         9.012249414223465 deg     Horiz. FPA:    -1.848322573870349e-06 deg
      Decl:         5.860936257179958 deg        Azimuth:         62.06917456622831 deg
       |R|:     6677.0095155241742759 km             |V|:        7.7264764301227755 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.8883261152955 deg       Mean Anom:         359.8883279636181 deg
 Long Peri:         10.58265675837723 deg        Arg. Lat:         12.36094790223884 deg
 True Long:         10.47098102533858 deg        Vert FPA:         90.00000184832257 deg
  Ang. Mom:         51589.75664540299 km^2/sec          p:     6677.1200270532126524 km
        C3:        -59.69646191409275 km^2/sec^2   Energy:        -29.84823095704638 km^2/sec^2
   Vel. RA:         102.1110547633917 deg       Vel. Decl:         27.77215591952032 deg
 Rad. Peri:     6677.0095153142592608 km        Vel. Peri:        7.7264764303656790 km/sec
  Rad. Apo:     6677.2305424504038456 km         Vel. Apo:         7.726220671492739 km/sec
 Mean Mot.:       0.06629911714028736 deg/sec
    Period:         5429.936559158828 sec          Period:         90.49894265264714 min
    Period:         1.508315710877452 hr           Period:       0.06284648795322718 day
               Time Past Periapsis:          5428.252192289423 sec
```

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          Time Past Ascending Node:           186.4359967267419 sec
   Beta Angle (Orbit plane to Sun):            5.12143720969015 deg
Mean Sidereal Greenwich Hour Angle:            96.5527697654835 deg

Geodetic Parameters:
  Latitude:         6.042742032728504 deg
 Longitude:        -87.20591004461249 deg
  Altitude:      299.1076081424806716 km
Geocentric Parameters:
  Latitude:         6.004384808544236 deg
 Longitude:        -87.20591004461249 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Finite
Name: Target Sequence.Maneuver
User Comment:  Maneuvers satellite with an impulsive burn or finite burn

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 01:08:36.391 [2461021.54764341]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    12
  Average step size:  63.5412 sec
  Largest step size:  68.214 sec
  Smallest step size:  54.474 sec

Maneuver Summary:
  Maneuver Start: 12 Dec 2025 00:55:53.489 UTCG;   2461021.53881353 UTC Julian Date
  Maneuver Stop:  12 Dec 2025 01:08:36.391 UTCG;   2461021.54764341 UTC Julian Date
  Duration:  762.902 sec
  Fuel Used:         129.657177136733 kg
  DeltaV Magnitude:        265.9685981883817 m/sec
  Maneuver Direction Specification:  Thrust Vector
  Maneuver direction is updated during maneuver.
  Burn centering is ON.
  Thrust Efficiency:  1  (Affects acceleration, but not mass flow rate)

Thrust vector at maneuver start with respect to VNC(Earth) axes:

       X (Velocity):        0.9987733657867121
         Y (Normal):      -0.03440844785408903
      Z (Co-Normal):       -0.0356064953534514
            Azimuth:        -1.973099725460614 deg
          Elevation:         -2.04053323431839 deg

Thrust vector at maneuver start with respect to Earth Inertial axes:

              X:        0.2183035790766011
              Y:        0.8721417977032243
              Z:        0.4378495541408421
        Azimuth:         75.94717899497783 deg
      Elevation:         25.96675452960473 deg

Thrust vector at maneuver stop with respect to VNC(Earth) axes:

       X (Velocity):        0.9987733657867119
         Y (Normal):      -0.03440844785408881
      Z (Co-Normal):      -0.03560649535345158
            Azimuth:        -1.973099725460601 deg
          Elevation:        -2.040533234318401 deg

Thrust vector at maneuver stop with respect to Earth Inertial axes:

              X:       -0.6034912247657971
              Y:         0.721293900156977
              Z:        0.3399021200684323
        Azimuth:         129.9185042045975 deg
      Elevation:         19.87091080997823 deg

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to VNC(Earth) axes:

       X (Velocity):         228.4503497674978 m/sec
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         Y (Normal):        -9.307959040585729 m/sec
      Z (Co-Normal):        -118.8556751082301 m/sec
            Azimuth:        -2.333163223216737 deg
          Elevation:        -27.46711906856027 deg
          Magnitude:         257.6875470716939 m/sec

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to Earth Inertial axes:

              X:        -57.15386177953872 m/sec
              Y:         225.5515487896843 m/sec
              Z:         110.7375583895837 m/sec
        Azimuth:         104.2192277864502 deg
      Elevation:         25.45079308912757 deg
      Magnitude:         257.6875470716938 m/sec

Thrust vector at maneuver start with respect to spacecraft body axes:

              X:                         1
              Y:                         0
              Z:      4.85722573273506e-17
        Azimuth:                         0 deg
      Elevation:                         0 deg

Thrust vector at maneuver stop with respect to spacecraft body axes:

              X:                         1
              Y:     5.551115123125783e-17
              Z:    -1.249000902703301e-16
        Azimuth:     3.180554681463517e-15 deg
      Elevation:                         0 deg

Attitude with respect to Earth Inertial axes:
  ---  ----Maneuver Start-----  -----Maneuver Stop-----
  qx:       0.6642516800741112       0.3760093857622278
  qy:       0.3282422249974282       0.4795717390769419
  qz:       0.5320763593606065       0.7561458585892117
  qs:       0.4097822532236129       0.2384771046365115

Data for Engine "Constant_Thrust_and_Isp":
  UserComment: Engine that has a constant Thrust and Isp
  Description: Engine that has a constant Thrust and Isp

  Engine values at beginning of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

  Engine values at end of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 12 Dec 2025 00:55:53.489  UTC Julian Date: 2461021.53881353
Julian Ephemeris Date: 2461021.53961428
Time past epoch: 21353.5 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6459.9073650101736348 km              Vx:        1.9513586852484399 km/sec
         Y:    -1527.0163616839122369 km              Vy:        6.5577036487738498 km/sec
         Z:     -712.7637609864340220 km              Vz:        3.5921396731618116 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.1101587077773729 km            RAAN:         358.1110819610446 deg
       ecc:        0.0006298500320251                  w:         271.0316646125914 deg
       inc:         28.49069717293417 deg             TA:         76.03509774091452 deg

Parameter Set Type:  Spherical
 Right Asc:         346.7003446584968 deg     Horiz. FPA:       0.03501580286480371 deg
      Decl:        -6.128785202029714 deg        Azimuth:         62.12219558192401 deg
       |R|:     6676.0927429401835980 km             |V|:        7.7275316431017895 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.00007927030646 deg       Mean Anom:         75.96506347002784 deg
 Long Peri:          269.142746573636 deg        Arg. Lat:          347.066762353506 deg
 True Long:         345.1778443145506 deg        Vert FPA:         89.96498419713519 deg
  Ang. Mom:         51589.70828914364 km^2/sec          p:     6677.1075098243109096 km
        C3:        -59.69655015803143 km^2/sec^2   Energy:        -29.84827507901571 km^2/sec^2
   Vel. RA:         73.42871180346603 deg       Vel. Decl:         27.70048617417489 deg
 Rad. Peri:     6672.9045806604799509 km        Vel. Peri:        7.7312222384629576 km/sec
  Rad. Apo:     6681.3157367550747949 km         Vel. Apo:         7.721489347575616 km/sec
 Mean Mot.:       0.06629926414642143 deg/sec
    Period:         5429.924519296967 sec           Period:         90.49874198828277 min
    Period:          1.50831236647138 hr            Period:       0.06284634860297415 day
              Time Past Periapsis:           1145.790446516262 sec
          Time Past Ascending Node:          5234.883007070227 sec
```

```
   Beta Angle (Orbit plane to Sun):            5.12145459059506 deg
Mean Sidereal Greenwich Hour Angle:             94.959040036378 deg

Geodetic Parameters:
  Latitude:        -6.026421822131796 deg
 Longitude:        -107.9225185908515 deg
  Altitude:      298.1895717623510791 km
Geocentric Parameters:
  Latitude:        -5.988161412599528 deg
 Longitude:        -107.9225185908515 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     5429.9245192969665368 sec

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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 12 Dec 2025 01:08:36.391  UTC Julian Date: 2461021.54764341
Julian Ephemeris Date: 2461021.54844415
Time past epoch: 22116.4 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5398.8582847665893496 km             Vx:       -4.5660086288995583 km/sec
         Y:     3491.7913402456124459 km             Vy:        5.7398479424612958 km/sec
         Z:     1986.7086929142526515 km             Vz:        3.0204783953193326 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2268760654187645 km           RAAN:          358.066995867309 deg
       ecc:        0.0674615973553220                 w:         14.11097523285365 deg
       inc:         28.41585003838438 deg            TA:         24.23314029763948 deg

Parameter Set Type:  Spherical
 Right Asc:          32.8933655520819 deg     Horiz. FPA:         1.494220290058728 deg
      Decl:         17.17074037335736 deg        Azimuth:         67.00609446687963 deg
       |R|:     6729.5831203209127125 km            |V|:        7.9320601950787815 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          22.6923725966655 deg       Mean Anom:         21.20121826367638 deg
 Long Peri:         12.17797110016264 deg        Arg. Lat:         38.34411553049313 deg
 True Long:         36.41111139780212 deg        Vert FPA:         88.50577970994127 deg
  Ang. Mom:         53361.30727320022 km^2/sec          p:     7143.5673859003754842 km
        C3:        -55.54457075896472 km^2/sec^2   Energy:        -27.77228537948236 km^2/sec^2
   Vel. RA:         128.5020017190874 deg       Vel. Decl:         22.38285371558078 deg
 Rad. Peri:     6692.1071480218524812 km        Vel. Peri:        7.9737676180175185 km/sec
  Rad. Apo:     7660.3466041089841383 km         Vel. Apo:          6.96591290589611 km/sec
 Mean Mot.:       0.05950415323779496 deg/sec
    Period:         6049.997864205226 sec          Period:         100.8332977367538 min
    Period:         1.680554962279229 hr           Period:       0.07002312342830122 day
             Time Past Periapsis:           356.2981255938637 sec
         Time Past Ascending Node:           563.258668058363 sec
   Beta Angle (Orbit plane to Sun):            5.05047380783783 deg
Mean Sidereal Greenwich Hour Angle:             98.146499494589 deg

Geodetic Parameters:
  Latitude:         17.39720540138068 deg
 Longitude:        -64.89675761480245 deg
  Altitude:      353.3439065590340533 km
Geocentric Parameters:
  Latitude:         17.29345348084724 deg
 Longitude:        -64.89675761480245 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.343 kg
     Total Mass:  1370.34 kg
Area/Mass Ratio:  1.45949e-08 km^2/kg
  Tank Pressure:  5000                      Pa
```

```
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     6049.9978642052265059 sec


***<<<< End of Sequence: Target Sequence >>>>***


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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 01:08:36.391  UTC Julian Date: 2461021.54764341
Julian Ephemeris Date: 2461021.54844415
Time past epoch: 22116.4 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5398.8582847665893496 km              Vx:       -4.5660086288995583 km/sec
         Y:     3491.7913402456124459 km              Vy:        5.7398479424612958 km/sec
         Z:     1986.7086929142526515 km              Vz:        3.0204783953193326 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2268760654187645 km            RAAN:          358.066995867309 deg
       ecc:        0.0674615973553220                  w:         14.11097523285365 deg
       inc:         28.41585003838438 deg             TA:         24.23314029763948 deg

Parameter Set Type:  Spherical
 Right Asc:          32.8933655520819 deg     Horiz. FPA:         1.494220290058728 deg
      Decl:         17.17074037335736 deg        Azimuth:         67.00609446687963 deg
       |R|:     6729.5831203209127125 km             |V|:        7.9320601950787815 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          22.6923725966655 deg       Mean Anom:         21.20121826367638 deg
 Long Peri:         12.17797110016264 deg        Arg. Lat:         38.34411553049313 deg
 True Long:         36.41111139780212 deg        Vert FPA:         88.50577970994127 deg
  Ang. Mom:         53361.30727320022 km^2/sec          p:     7143.5673859003754842 km
        C3:        -55.54457075896472 km^2/sec^2   Energy:        -27.77228537948236 km^2/sec^2
   Vel. RA:         128.5020017190874 deg       Vel. Decl:         22.38285371558078 deg
 Rad. Peri:     6692.1071480218524812 km        Vel. Peri:        7.9737676180175185 km/sec
  Rad. Apo:     7660.3466041089841383 km         Vel. Apo:          6.96591290589611 km/sec
 Mean Mot.:       0.05950415323779496 deg/sec
    Period:         6049.997864205226 sec          Period:         100.8332977367538 min
    Period:         1.680554962279229 hr           Period:       0.07002312342830122 day
              Time Past Periapsis:           356.2981255938637 sec
          Time Past Ascending Node:           563.258668058363 sec
   Beta Angle (Orbit plane to Sun):           5.05047380783783 deg
Mean Sidereal Greenwich Hour Angle:            98.146499494589 deg

Geodetic Parameters:
  Latitude:         17.39720540138068 deg
 Longitude:        -64.89675761480245 deg
  Altitude:      353.3439065590340533 km
Geocentric Parameters:
  Latitude:         17.29345348084724 deg
 Longitude:        -64.89675761480245 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.343 kg
     Total Mass:  1370.34 kg
Area/Mass Ratio:  1.45949e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate1
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  19 Dec 2025 01:08:36.391 [2461028.54764341]:  Stopped on: Duration; Run Sequence STOP
```

```
Propagation Statistics:
  Number of steps:    8185
  Average step size:  73.8904 sec
  Largest step size:  83.138 sec
  Smallest step size:  39.603 sec

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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 19 Dec 2025 01:08:36.391  UTC Julian Date: 2461028.54764341
Julian Ephemeris Date: 2461028.54844415
Time past epoch: 626916 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -2278.3091637843390345 km             Vx:       -6.9450378508394826 km/sec
         Y:     6374.1407336290212697 km             Vy:       -1.1837277063124581 km/sec
         Z:     1658.9614010884943127 km             Vz:       -3.0306788967113136 km/sec

Parameter Set Type:  Keplerian
       sma:     7173.4441229842650500 km           RAAN:         316.7163427097117 deg
       ecc:        0.0671039680669213                 w:         81.43064361589701 deg
       inc:          28.3224206632065 deg            TA:         68.45511873139301 deg

Parameter Set Type:  Spherical
 Right Asc:          109.668414039999 deg     Horiz. FPA:         3.485827392353009 deg
      Decl:         13.77059873578929 deg       Azimuth:         114.9951993117994 deg
       |R|:     6969.3985155244145062 km            |V|:        7.6694052316497237 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          64.9196895608401 deg       Mean Anom:         61.43742177647216 deg
 Long Peri:         38.14698632560876 deg        Arg. Lat:           149.88576234729 deg
 True Long:         106.6021050570018 deg        Vert FPA:         86.51417260764698 deg
  Ang. Mom:          53352.2498104167 km^2/sec          p:     7141.1425163539588539 km
        C3:         -55.5661178460781 km^2/sec^2   Energy:        -27.78305892303905 km^2/sec^2
   Vel. RA:         189.6726694703974 deg       Vel. Decl:        -23.27625189647986 deg
 Rad. Peri:     6692.0775576256846762 km        Vel. Peri:        7.9724494151478140 km/sec
  Rad. Apo:     7654.8106883428454239 km         Vel. Apo:         6.969767376698729 km/sec
 Mean Mot.:       0.05953878125340342 deg/sec
    Period:         6046.479158983815 sec          Period:         100.7746526497303 min
    Period:         1.679577544162171 hr           Period:       0.06998239767342378 day
              Time Past Periapsis:          1031.889139869826 sec
         Time Past Ascending Node:          2272.913973941585 sec
   Beta Angle (Orbit plane to Sun):         -0.929036304482487 deg
Mean Sidereal Greenwich Hour Angle:           105.046031097185 deg

Geodetic Parameters:
  Latitude:         13.80479309999648 deg
 Longitude:          4.98863771618307 deg
  Altitude:      592.4701439012966375 km
Geocentric Parameters:
  Latitude:         13.72343851831034 deg
 Longitude:          4.98863771618307 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.343 kg
     Total Mass:  1370.34 kg
Area/Mass Ratio:  1.45949e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


                              Valk2 - Astrogator Mission Control Sequence Summary
-------------------------------------------------------------------------------------------------------------------

***-------------------------------------------------------------------------------***
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)
```

```
State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
              Time Past Periapsis:            1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
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```
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
              Time Past Periapsis:           1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 03:19:00.910 [2461021.63820497]:  Stopped on: Periapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    210
  Average step size:  65.9465 sec
  Largest step size:  67.172 sec
  Smallest step size:  41.067 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 03:19:00.910  UTC Julian Date: 2461021.63820497
Julian Ephemeris Date: 2461021.63900571
Time past epoch: 29940.9 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6398.2989481569957206 km              Vx:        2.1451929664865044 km/sec
         Y:    -1556.2211716765382334 km              Vy:       -6.5647327892499714 km/sec
         Z:    -1000.9995879988719025 km              Vz:       -3.5059052476385379 km/sec

Parameter Set Type:  Keplerian
       sma:     6676.4800976577607798 km            RAAN:         357.4018670779532 deg
       ecc:        0.0023958111149361                  w:         198.3671875132011 deg
       inc:         28.48635841831204 deg             TA:         359.999981604985 deg

Parameter Set Type:  Spherical
 Right Asc:          193.670277223415 deg     Horiz. FPA:     -4.39656470283061e-08 deg
      Decl:        -8.643692570170352 deg        Azimuth:         117.2486132842139 deg
       |R|:     6660.4845124311441396 km             |V|:        7.7452528082133192 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.9999816289812 deg       Mean Anom:         359.9999816729947 deg
 Long Peri:         195.7690545911543 deg        Arg. Lat:         198.3671691181862 deg
 True Long:         195.7690361961394 deg        Vert FPA:         90.00000004396564 deg
  Ang. Mom:         51587.13637396863 km^2/sec          p:     6676.4417752568842843 km
        C3:        -59.70218373598341 km^2/sec^2   Energy:        -29.85109186799171 km^2/sec^2
   Vel. RA:         288.0961167366871 deg       Vel. Decl:        -26.91397015673316 deg
 Rad. Peri:     6660.4845124311423206 km        Vel. Peri:        7.7452528082133210 km/sec
  Rad. Apo:     6692.4756828843774201 km         Vel. Apo:         7.708229184291215 km/sec
 Mean Mot.:       0.06630864938434901 deg/sec
    Period:         5429.155975011786 sec          Period:         90.48593291686309 min
    Period:         1.508098881947718 hr           Period:       0.06283745341448826 day
```

```
              Time Past Periapsis:          5429.155698622424 sec
          Time Past Ascending Node:         2992.879930001726 sec
   Beta Angle (Orbit plane to Sun):          5.16418681610098 deg
Mean Sidereal Greenwich Hour Angle:          130.837922650162 deg

Geodetic Parameters:
  Latitude:        -8.840758049621734 deg
 Longitude:         63.16977283932487 deg
  Altitude:      282.8486402853465052 km
Geocentric Parameters:
  Latitude:        -8.784974392370813 deg
 Longitude:         63.16977283932487 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: TargeterSequence
Name: Target Sequence
User Comment:  Sequence that runs targeting profiles

Sequence Start: 12 Dec 2025 03:19:00.910 UTCG;   2461021.63820497 UTC Julian Date
Sequence Stop:  12 Dec 2025 04:08:48.797 UTCG;   2461021.672787 UTC Julian Date

***<<<< Start of Sequence: Target Sequence >>>>***


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Target Sequence.Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 04:02:27.024 [2461021.66836833]:  Stopped on: Apoapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    40
  Average step size:  66.506 sec
  Largest step size:  67.164 sec
  Smallest step size:  60 sec

----------------------------------
Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 04:02:27.024  UTC Julian Date: 2461021.66836833
Julian Ephemeris Date: 2461021.66916907
Time past epoch: 32547 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6586.9178894995102382 km             Vx:       -1.2479371590721435 km/sec
         Y:      874.2120956207788822 km             Vy:        6.7178740157499206 km/sec
         Z:      650.5783995731553659 km             Vz:        3.6078862627221553 km/sec

Parameter Set Type:  Keplerian
       sma:     6676.6257055830083118 km           RAAN:         357.1664098313416 deg
       ecc:        0.0000262972020386                 w:         11.78816168367132 deg
       inc:         28.48891017896323 deg            TA:         359.9997468140807 deg

Parameter Set Type:  Spherical
 Right Asc:         7.560081839393122 deg     Horiz. FPA:    -6.657906003582483e-09 deg
      Decl:         5.591989665342763 deg        Azimuth:         62.02009014881016 deg
       |R|:     6676.4501290078942475 km            |V|:        7.7268377574024587 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          359.999746821922 deg       Mean Anom:         359.9997468285799 deg
 Long Peri:         8.954571515012956 deg        Arg. Lat:         11.78790849775208 deg
 True Long:         8.954318329093701 deg        Vert FPA:          90.0000000066579 deg
  Ang. Mom:         51587.84694223271 km^2/sec          p:     6676.6257009658356765 km
        C3:        -59.70088171435004 km^2/sec^2   Energy:        -29.85044085717502 km^2/sec^2
   Vel. RA:         100.5235213391396 deg       Vel. Decl:         27.83514893272304 deg
 Rad. Peri:     6676.4501290078915190 km        Vel. Peri:        7.7268377574024614 km/sec
```

```
  Rad. Apo:     6676.8012821581232856 km         Vel. Apo:         7.726431379661807 km/sec
 Mean Mot.:       0.06630648024697765 deg/sec
    Period:         5429.333583370373 sec          Period:         90.48889305617288 min
    Period:         1.508148217602882 hr           Period:       0.06283950906678672 day
              Time Past Periapsis:           5429.329765170114 sec
         Time Past Ascending Node:           177.7698477384074 sec
   Beta Angle (Orbit plane to Sun):           5.18124138826472 deg
Mean Sidereal Greenwich Hour Angle:           141.726461616095 deg

Geodetic Parameters:
  Latitude:         5.772590826551371 deg
 Longitude:        -133.8322250969053 deg
  Altitude:      298.5277388684558559 km
Geocentric Parameters:
  Latitude:          5.73592164750634 deg
 Longitude:        -133.8322250969053 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Finite
Name: Target Sequence.Maneuver
User Comment:  Maneuvers satellite with an impulsive burn or finite burn

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 04:08:48.797 [2461021.672787]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    11
  Average step size:  64.9706 sec
  Largest step size:  67.318 sec
  Smallest step size:  60 sec

Maneuver Summary:
  Maneuver Start: 12 Dec 2025 03:56:05.251 UTCG;   2461021.66394966 UTC Julian Date
  Maneuver Stop:  12 Dec 2025 04:08:48.797 UTCG;   2461021.672787 UTC Julian Date
  Duration:  763.546 sec
  Fuel Used:          129.76674971286 kg
  DeltaV Magnitude:        266.2038494367835 m/sec
  Maneuver Direction Specification:  Thrust Vector
  Maneuver direction is updated during maneuver.
  Burn centering is ON.
  Thrust Efficiency:  1  (Affects acceleration, but not mass flow rate)

Thrust vector at maneuver start with respect to VNC(Earth) axes:

       X (Velocity):        0.9987597200090511
         Y (Normal):       -0.0346231603788929
      Z (Co-Normal):      -0.03578069944564649
           Azimuth:        -1.985429367384609 deg
         Elevation:        -2.050520757832222 deg

Thrust vector at maneuver start with respect to Earth Inertial axes:

              X:        0.2425963957472508
              Y:        0.8662931516242134
              Z:        0.4366728343043917
        Azimuth:         74.35564910642815 deg
      Elevation:         25.89178677941168 deg

Thrust vector at maneuver stop with respect to VNC(Earth) axes:

       X (Velocity):         0.998759720009051
         Y (Normal):      -0.03462316037889268
      Z (Co-Normal):      -0.03578069944564631
           Azimuth:        -1.985429367384597 deg
         Elevation:        -2.050520757832212 deg

Thrust vector at maneuver stop with respect to Earth Inertial axes:

              X:       -0.5839073630102009
              Y:        0.7360373956964438
              Z:        0.3424925452602875
        Azimuth:          128.425392882349 deg
```

```
      Elevation:         20.02880635043971 deg

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to VNC(Earth) axes:

       X (Velocity):         228.5645481097947 m/sec
         Y (Normal):        -9.372341741758845 m/sec
      Z (Co-Normal):        -119.0916920869129 m/sec
            Azimuth:         -2.34811111503967 deg
          Elevation:        -27.50168103886623 deg
          Magnitude:         257.9000282405627 m/sec

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to Earth Inertial axes:

              X:        -50.99568646633778 m/sec
              Y:         227.1036364057971 m/sec
              Z:         111.0666595318932 m/sec
        Azimuth:         102.6557359025294 deg
      Elevation:         25.50931320991197 deg
      Magnitude:         257.9000282405626 m/sec

Thrust vector at maneuver start with respect to spacecraft body axes:

              X:        0.9999999999999999
              Y:    -5.551115123125783e-17
              Z:    -3.191891195797325e-16
        Azimuth:    -3.180554681463518e-15 deg
      Elevation:                         0 deg

Thrust vector at maneuver stop with respect to spacecraft body axes:

              X:                         1
              Y:     5.551115123125783e-17
              Z:     1.387778780781446e-16
        Azimuth:     3.180554681463517e-15 deg
      Elevation:                         0 deg

Attitude with respect to Earth Inertial axes:
  ---  ----Maneuver Start-----  -----Maneuver Stop-----
  qx:        0.671176722829759       0.3845889756761377
  qy:       0.3226769948054146       0.4784571596224464
  qz:       0.5240051136675299       0.7503548679865557
  qs:       0.4133037679541893       0.2452297663075972

Data for Engine "Constant_Thrust_and_Isp":
  UserComment: Engine that has a constant Thrust and Isp
  Description: Engine that has a constant Thrust and Isp

  Engine values at beginning of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

  Engine values at end of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 12 Dec 2025 03:56:05.251  UTC Julian Date: 2461021.66394966
Julian Ephemeris Date: 2461021.6647504
Time past epoch: 32165.3 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6414.4792704884785053 km              Vx:        2.1367580249404607 km/sec
         Y:    -1691.8120641301134128 km              Vy:        6.5050205274611654 km/sec
         Z:     -744.9960922021954275 km              Vz:        3.5832437669499964 km/sec

Parameter Set Type:  Keplerian
       sma:     6676.6597482875713467 km            RAAN:         357.1675150785737 deg
       ecc:        0.0006317640011119                  w:         271.8736977253949 deg
       inc:         28.48853883592171 deg             TA:         74.59523543120855 deg

Parameter Set Type:  Spherical
 Right Asc:         345.2247710325245 deg     Horiz. FPA:       0.03489109832038449 deg
      Decl:        -6.407610204827685 deg        Azimuth:         62.18238310421508 deg
       |R|:     6675.5367985847497039 km             |V|:        7.7279145190164078 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         74.56034139871514 deg       Mean Anom:         74.52545030304475 deg
 Long Peri:         269.0412128039687 deg        Arg. Lat:         346.4689331566035 deg
 True Long:         343.6364482351772 deg        Vert FPA:         89.96510890167961 deg
  Ang. Mom:         51587.96818262515 km^2/sec          p:     6676.6570834607209690 km
        C3:        -59.70057731371333 km^2/sec^2   Energy:        -29.85028865685667 km^2/sec^2
   Vel. RA:         71.81577085371893 deg       Vel. Decl:         27.62452910568713 deg
 Rad. Peri:     6672.4416750109294298 km        Vel. Peri:        7.7314978077407703 km/sec
  Rad. Apo:     6680.8778215642123541 km         Vel. Apo:         7.721735011544745 km/sec
```

```
 Mean Mot.:       0.06630597312609859 deg/sec
    Period:         5429.375107961441 sec          Period:         90.48958513269069 min
    Period:         1.508159752211512 hr           Period:       0.06283998967547964 day
               Time Past Periapsis:           1123.962846624913 sec
           Time Past Ascending Node:           5225.34376138822 sec
   Beta Angle (Orbit plane to Sun):             5.1808414385888 deg
Mean Sidereal Greenwich Hour Angle:            140.131385093495 deg

Geodetic Parameters:
  Latitude:        -6.307960458639807 deg
 Longitude:        -154.5698584739543 deg
  Altitude:      297.6558971303907697 km
Geocentric Parameters:
  Latitude:        -6.267937469093356 deg
 Longitude:        -154.5698584739543 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     5429.3751079614412447 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 04:08:48.797  UTC Julian Date: 2461021.672787
Julian Ephemeris Date: 2461021.67358774
Time past epoch: 32928.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5492.6670544428889116 km              Vx:       -4.4080044358657320 km/sec
         Y:     3355.8399213683173912 km              Vy:        5.8514928825428640 km/sec
         Z:     1962.3019798569387149 km              Vz:        3.0421191684849918 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2330968569667675 km            RAAN:         357.1250372335027 deg
       ecc:        0.0675346647262556                  w:         13.54025735078425 deg
       inc:         28.41382688091461 deg             TA:         24.25485321895042 deg

Parameter Set Type:  Spherical
 Right Asc:         31.42356564411157 deg     Horiz. FPA:          1.49701753560927 deg
      Decl:         16.95445561177066 deg        Azimuth:         66.85280871283921 deg
       |R|:     6729.1665166617849536 km            |V|:        7.9325255181641614 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.71115693085438 deg       Mean Anom:         21.21721722397604 deg
 Long Peri:         10.66529458428694 deg        Arg. Lat:         37.79511056973467 deg
 True Long:         34.92014780323734 deg        Vert FPA:         88.50298246439073 deg
  Ang. Mom:         53361.06602443311 km^2/sec          p:     7143.5027932950997638 km
        C3:        -55.54452260958166 km^2/sec^2   Energy:        -27.77226130479083 km^2/sec^2
   Vel. RA:         126.9911823343993 deg       Vel. Decl:          22.5506157910441 deg
 Rad. Peri:     6691.5886006632727003 km        Vel. Peri:        7.9743494719839694 km/sec
  Rad. Apo:     7660.8775930506608347 km         Vel. Apo:         6.965398595173748 km/sec
 Mean Mot.:       0.05950407586512988 deg/sec
    Period:         6050.005730968161 sec          Period:         100.8334288494693 min
    Period:         1.680557147491156 hr           Period:       0.07002321447879815 day
               Time Past Periapsis:           356.5674605562539 sec
           Time Past Ascending Node:           555.1078194929366 sec
   Beta Angle (Orbit plane to Sun):             5.10928327071684 deg
Mean Sidereal Greenwich Hour Angle:            143.321538306792 deg

Geodetic Parameters:
  Latitude:         17.18174091643876 deg
 Longitude:        -111.5428806548399 deg
  Altitude:      352.8819628822118943 km
Geocentric Parameters:
  Latitude:         17.07910928481641 deg
 Longitude:        -111.5428806548399 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
```

```
       Dry Mass:  500 kg
      Fuel Mass:  870.233 kg
     Total Mass:  1370.23 kg
Area/Mass Ratio:  1.45961e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     6050.0057309681615152 sec


***<<<< End of Sequence: Target Sequence >>>>***


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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 04:08:48.797  UTC Julian Date: 2461021.672787
Julian Ephemeris Date: 2461021.67358774
Time past epoch: 32928.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5492.6670544428889116 km             Vx:       -4.4080044358657320 km/sec
         Y:     3355.8399213683173912 km             Vy:        5.8514928825428640 km/sec
         Z:     1962.3019798569387149 km             Vz:        3.0421191684849918 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2330968569667675 km           RAAN:         357.1250372335027 deg
       ecc:        0.0675346647262556                 w:         13.54025735078425 deg
       inc:         28.41382688091461 deg            TA:         24.25485321895042 deg

Parameter Set Type:  Spherical
 Right Asc:         31.42356564411157 deg     Horiz. FPA:          1.49701753560927 deg
      Decl:         16.95445561177066 deg        Azimuth:         66.85280871283921 deg
       |R|:     6729.1665166617849536 km            |V|:        7.9325255181641614 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.71115693085438 deg       Mean Anom:         21.21721722397604 deg
 Long Peri:         10.66529458428694 deg        Arg. Lat:         37.79511056973467 deg
 True Long:         34.92014780323734 deg        Vert FPA:         88.50298246439073 deg
  Ang. Mom:         53361.06602443311 km^2/sec          p:     7143.5027932950997638 km
        C3:        -55.54452260958166 km^2/sec^2   Energy:        -27.77226130479083 km^2/sec^2
   Vel. RA:         126.9911823343993 deg       Vel. Decl:          22.5506157910441 deg
 Rad. Peri:     6691.5886006632727003 km        Vel. Peri:        7.9743494719839694 km/sec
  Rad. Apo:     7660.8775930506608347 km         Vel. Apo:         6.965398595173748 km/sec
 Mean Mot.:       0.05950407586512988 deg/sec
    Period:         6050.005730968161 sec          Period:         100.8334288494693 min
    Period:         1.680557147491156 hr           Period:       0.07002321447879815 day
              Time Past Periapsis:          356.5674605562539 sec
         Time Past Ascending Node:          555.1078194929366 sec
   Beta Angle (Orbit plane to Sun):          5.10928327071684 deg
Mean Sidereal Greenwich Hour Angle:          143.321538306792 deg

Geodetic Parameters:
  Latitude:         17.18174091643876 deg
 Longitude:        -111.5428806548399 deg
  Altitude:      352.8819628822118943 km
Geocentric Parameters:
  Latitude:         17.07910928481641 deg
 Longitude:        -111.5428806548399 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.233 kg
     Total Mass:  1370.23 kg
Area/Mass Ratio:  1.45961e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate1
User Comment:  Propagates until stopping conditions are met
```

```
Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  19 Dec 2025 04:08:48.797 [2461028.672787]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    8184
  Average step size:  73.8933 sec
  Largest step size:  83.15 sec
  Smallest step size:  40.871 sec

----------------------------------
Satellite State at End of Segment:
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UTC Gregorian Date: 19 Dec 2025 04:08:48.797  UTC Julian Date: 2461028.672787
Julian Ephemeris Date: 2461028.67358774
Time past epoch: 637729 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -2186.0338803533027203 km              Vx:       -6.9560132271804216 km/sec
         Y:     6412.6598798979312051 km              Vy:       -1.0789534947003938 km/sec
         Z:     1654.7085491528464445 km              Vz:       -3.0314577062589652 km/sec

Parameter Set Type:  Keplerian
       sma:     7173.2422834944100032 km            RAAN:         315.7726558608628 deg
       ecc:        0.0671721225986206                  w:         80.83835355178236 deg
       inc:         28.32127499832396 deg             TA:         69.15398107691961 deg

Parameter Set Type:  Spherical
 Right Asc:          108.823908205273 deg     Horiz. FPA:         3.508385343327276 deg
      Decl:         13.72502830229531 deg        Azimuth:         115.0177780052557 deg
       |R|:     6974.1674229935570111 km            |V|:        7.6642022732487973 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         65.59688804047998 deg       Mean Anom:         62.09204520414509 deg
 Long Peri:         36.61100941264517 deg        Arg. Lat:          149.992334628702 deg
 True Long:         105.7649904895648 deg        Vert FPA:         86.49161465667272 deg
  Ang. Mom:         53351.25398934629 km^2/sec          p:     7140.8759396362547704 km
        C3:        -55.56768135619473 km^2/sec^2   Energy:        -27.78384067809736 km^2/sec^2
   Vel. RA:         188.8169380899421 deg       Vel. Decl:        -23.29932414282007 deg
 Rad. Peri:     6691.4003733979143362 km        Vel. Peri:        7.9731074232902834 km/sec
  Rad. Apo:     7655.0841935909056701 km         Vel. Apo:         6.969388270610239 km/sec
 Mean Mot.:       0.05954129420946173 deg/sec
    Period:         6046.223965732882 sec          Period:         100.7703994288814 min
    Period:         1.679506657148023 hr           Period:       0.06997944404783428 day
             Time Past Periapsis:           1042.840032762976 sec
          Time Past Ascending Node:         2274.006125081123 sec
   Beta Angle (Orbit plane to Sun):         -1.23919075634662 deg
Mean Sidereal Greenwich Hour Angle:          150.221069909389 deg

Geodetic Parameters:
  Latitude:         13.76095420264134 deg
 Longitude:        -41.03084880964259 deg
  Altitude:      597.2315366494167392 km
Geocentric Parameters:
  Latitude:         13.67989331788617 deg
 Longitude:        -41.03084880964259 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.233 kg
     Total Mass:  1370.23 kg
Area/Mass Ratio:  1.45961e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000

                             Valk3 - Astrogator Mission Control Sequence Summary
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***-------------------------------------------------------------------------***
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

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Satellite State at Beginning of Segment:
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UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km             Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km             Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km             Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km           RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                 w:         283.0818133205355 deg
       inc:         28.49586471175197 deg            TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km            |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
              Time Past Periapsis:           1159.77625053902 sec
         Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
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UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km             Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km             Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km             Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km           RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                 w:         283.0818133205355 deg
       inc:         28.49586471175197 deg            TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km            |V|:        7.7266134375937217 km/sec
```

```
Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
              Time Past Periapsis:           1159.77625053902 sec
         Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 07:49:38.972 [2461021.82614551]:  Stopped on: Periapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    454
  Average step size:  66.2222 sec
  Largest step size:  67.184 sec
  Smallest step size:  41.067 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 07:49:38.972  UTC Julian Date: 2461021.82614551
Julian Ephemeris Date: 2461021.82694625
Time past epoch: 46179 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6416.6525964189486331 km              Vx:        2.0586611018425547 km/sec
         Y:    -1455.7615514774047369 km              Vy:       -6.5989858824376286 km/sec
         Z:    -1031.3115320304043507 km              Vz:       -3.4937679224033440 km/sec

Parameter Set Type:  Keplerian
       sma:     6675.8514167469202221 km            RAAN:         355.9920677592895 deg
       ecc:        0.0023668423209720                  w:         198.9468715675601 deg
       inc:         28.48418408506191 deg             TA:         359.9999862281963 deg

Parameter Set Type:  Spherical
 Right Asc:         192.7824551936898 deg     Horiz. FPA:    -3.251872895196909e-08 deg
      Decl:        -8.908120159466501 deg        Azimuth:         117.1669142240984 deg
       |R|:     6660.0507290852392543 km             |V|:        7.7453931149080679 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.9999862598135 deg       Mean Anom:         359.9999862923343 deg
 Long Peri:         194.9389393268497 deg        Arg. Lat:         198.9468577957564 deg
 True Long:          194.938925555046 deg        Vert FPA:         90.00000003251873 deg
  Ang. Mom:         51584.71106199528 km^2/sec          p:     6675.8140190106614682 km
        C3:        -59.70780603356122 km^2/sec^2   Energy:        -29.85390301678061 km^2/sec^2
   Vel. RA:         287.3261613256481 deg       Vel. Decl:        -26.81279780841238 deg
```

```
 Rad. Peri:     6660.0507290852419828 km        Vel. Peri:        7.7453931149080661 km/sec
  Rad. Apo:     6691.6521044085975518 km         Vel. Apo:         7.708815440063033 km/sec
 Mean Mot.:       0.06631801627126301 deg/sec
    Period:         5428.389150355144 sec          Period:         90.47315250591906 min
    Period:         1.507885875098651 hr           Period:       0.06282857812911045 day
             Time Past Periapsis:           5428.38894365919 sec
         Time Past Ascending Node:          3001.22168803148 sec
   Beta Angle (Orbit plane to Sun):         5.24073850882914 deg
Mean Sidereal Greenwich Hour Angle:         198.681760578679 deg

Geodetic Parameters:
  Latitude:        -9.107302535328611 deg
 Longitude:        -5.562073609898113 deg
  Altitude:      282.4452772373958851 km
Geocentric Parameters:
  Latitude:         -9.04988918315386 deg
 Longitude:        -5.562073609898113 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                     Pa
   Fuel Density:  1000                     kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: TargeterSequence
Name: Target Sequence
User Comment:  Sequence that runs targeting profiles

Sequence Start: 12 Dec 2025 07:49:38.972 UTCG;   2461021.82614551 UTC Julian Date
Sequence Stop:  12 Dec 2025 08:39:29.278 UTCG;   2461021.86075553 UTC Julian Date

***<<<< Start of Sequence: Target Sequence >>>>***


***-----------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Target Sequence.Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 08:33:06.989 [2461021.85633089]:  Stopped on: Apoapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    40
  Average step size:  66.4993 sec
  Largest step size:  67.038 sec
  Smallest step size:  60 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 08:33:06.989  UTC Julian Date: 2461021.85633089
Julian Ephemeris Date: 2461021.85713163
Time past epoch: 48787 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6592.8329227518197513 km             Vx:      -1.1805914250873815 km/sec
         Y:      786.7728903710145687 km             Vy:       6.7363019981579457 km/sec
         Z:      690.2414272216840345 km             Vz:       3.5980196460678093 km/sec

Parameter Set Type:  Keplerian
       sma:     6675.9995676046301014 km           RAAN:         355.7604321338568 deg
       ecc:        0.0000906441927551                 w:         12.52115526534074 deg
       inc:          28.4859972416688 deg            TA:         359.9999208567984 deg

Parameter Set Type:  Spherical
 Right Asc:         6.805356798339402 deg     Horiz. FPA:    -7.173219473073202e-09 deg
      Decl:         5.935040235026022 deg        Azimuth:         62.08833942069318 deg
       |R|:     6675.3944270129904908 km            |V|:        7.7276973347088660 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.9999208657292 deg       Mean Anom:         359.9999208729022 deg
 Long Peri:          8.28158739919756 deg        Arg. Lat:         12.52107612213917 deg
```

```
 True Long:         8.281508255995927 deg        Vert FPA:         90.00000000717321 deg
  Ang. Mom:          51585.4277217587 km^2/sec          p:     6675.999512752147893 5 km
        C3:        -59.70648102408718 km^2/sec^2   Energy:        -29.85324051204359 km^2/sec^2
   Vel. RA:         99.94059296266943 deg       Vel. Decl:          27.7490914901693 deg
 Rad. Peri:     6675.3944270129914003 km        Vel. Peri:        7.7276973347088651 km/sec
  Rad. Apo:     6676.6047081962688026 km         Vel. Apo:         7.726296519911071 km/sec
 Mean Mot.:       0.06631580873296464 deg/sec
    Period:         5428.569852018545 sec          Period:         90.47616420030909 min
    Period:         1.507936070005151 hr           Period:       0.06283066958354798 day
              Time Past Periapsis:           5428.568658832497 sec
          Time Past Ascending Node:          188.7758682524045 sec
   Beta Angle (Orbit plane to Sun):           5.25360493723304 deg
Mean Sidereal Greenwich Hour Angle:           209.578247677014 deg

Geodetic Parameters:
  Latitude:         6.118025279844692 deg
 Longitude:           157.56117482148 deg
  Altitude:      297.4983916537599953 km
Geocentric Parameters:
  Latitude:         6.079187994745323 deg
 Longitude:           157.56117482148 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Finite
Name: Target Sequence.Maneuver
User Comment:  Maneuvers satellite with an impulsive burn or finite burn

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 08:39:29.278 [2461021.86075553]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    12
  Average step size:  61.2906 sec
  Largest step size:  66.46 sec
  Smallest step size:  43.86 sec

Maneuver Summary:
  Maneuver Start: 12 Dec 2025 08:26:44.700 UTCG;   2461021.85190625 UTC Julian Date
  Maneuver Stop:  12 Dec 2025 08:39:29.278 UTCG;   2461021.86075553 UTC Julian Date
  Duration:  764.578 sec
  Fuel Used:        129.9421050085446 kg
  DeltaV Magnitude:        266.5803746864604 m/sec
  Maneuver Direction Specification:  Thrust Vector
  Maneuver direction is updated during maneuver.
  Burn centering is ON.
  Thrust Efficiency:  1  (Affects acceleration, but not mass flow rate)

Thrust vector at maneuver start with respect to VNC(Earth) axes:

       X (Velocity):        0.9987206112563688
         Y (Normal):       -0.0352480329131753
      Z (Co-Normal):      -0.03625902408307342
            Azimuth:        -2.021311662734586 deg
          Elevation:        -2.077944537053666 deg

Thrust vector at maneuver start with respect to Earth Inertial axes:

              X:         0.251657985190191
              Y:        0.8634188658621316
              Z:        0.4372369158286661
        Azimuth:         73.75034643863005 deg
      Elevation:         25.92771794895023 deg

Thrust vector at maneuver stop with respect to VNC(Earth) axes:

       X (Velocity):        0.9987206112563687
         Y (Normal):      -0.03524803291317519
      Z (Co-Normal):       -0.0362590240830734
            Azimuth:         -2.02131166273458 deg
          Elevation:        -2.077944537053666 deg

Thrust vector at maneuver stop with respect to Earth Inertial axes:
```

```
             X:       -0.5770411130956354
             Y:        0.7436183054984059
             Z:        0.3377208455589164
       Azimuth:         127.8111410333661 deg
     Elevation:         19.73807635868848 deg

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to VNC(Earth) axes:

      X (Velocity):         228.7171254604728 m/sec
        Y (Normal):        -9.557208345667775 m/sec
     Z (Co-Normal):        -119.5185075599903 m/sec
           Azimuth:        -2.392778080904944 deg
         Elevation:        -27.56927563783962 deg
         Magnitude:         258.2393025076385 m/sec

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to Earth Inertial axes:

             X:        -48.82899115037208 m/sec
             Y:          228.169205352755 m/sec
             Z:         110.6439366235711 m/sec
       Azimuth:          102.079292209739 deg
     Elevation:          25.3695544498956 deg
     Magnitude:         258.2393025076385 m/sec

Thrust vector at maneuver start with respect to spacecraft body axes:

             X:                         1
             Y:     1.665334536937735e-16
             Z:     1.387778780781446e-17
       Azimuth:     9.541664044390549e-15 deg
     Elevation:                         0 deg

Thrust vector at maneuver stop with respect to spacecraft body axes:

             X:                         1
             Y:                         0
             Z:     1.249000902703301e-16
       Azimuth:                         0 deg
     Elevation:                         0 deg

Attitude with respect to Earth Inertial axes:
  ---  ----Maneuver Start-----  -----Maneuver Stop-----
  qx:       0.6740044545116752       0.3866061064605146
  qy:       0.3202571066417097       0.4808630108732868
  qz:       0.5211587023646301       0.7465194714953489
  qs:       0.4141823123861214       0.2490284359257463

Data for Engine "Constant_Thrust_and_Isp":
  UserComment: Engine that has a constant Thrust and Isp
  Description: Engine that has a constant Thrust and Isp

  Engine values at beginning of segment:
           Thrust:                      500 N
              Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

  Engine values at end of segment:
           Thrust:                      500 N
              Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

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Satellite State at Beginning of Segment:
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UTC Gregorian Date: 12 Dec 2025 08:26:44.700  UTC Julian Date: 2461021.85190625
Julian Ephemeris Date: 2461021.85270699
Time past epoch: 48404.7 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6393.4910150721489117 km              Vx:        2.2066170036717017 km/sec
         Y:    -1780.9857947834163951 km              Vy:        6.4769558972952144 km/sec
         Z:     -707.3820313880978574 km              Vz:        3.5935635324251209 km/sec

Parameter Set Type:  Keplerian
       sma:     6676.0676075818082609 km            RAAN:         355.7615222212722 deg
       ecc:        0.0006376052246377                  w:         278.6526589396581 deg
       inc:         28.48618368959147 deg             TA:         68.50843038006438 deg

Parameter Set Type:  Spherical
 Right Asc:         344.4341656053861 deg     Horiz. FPA:       0.03398412394624216 deg
      Decl:        -6.083786106166342 deg        Azimuth:         62.11772592767619 deg
       |R|:     6674.5057569350683480 km             |V|:        7.7287654355644699 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          68.4744422808642 deg       Mean Anom:          68.4404581598406 deg
 Long Peri:         274.4141811609302 deg        Arg. Lat:         347.1610893197225 deg
 True Long:         342.9226115409946 deg        Vert FPA:         89.96601587605376 deg
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  Ang. Mom:         51585.68031952228 km^2/sec          p:     6676.0648934904629641 km
        C3:        -59.70587251802566 km^2/sec^2   Energy:        -29.85293625901283 km^2/sec^2
   Vel. RA:          71.1866722655726 deg       Vel. Decl:         27.70760618124902 deg
 Rad. Peri:     6671.8109119951786852 km        Vel. Peri:        7.7318858402861705 km/sec
  Rad. Apo:     6680.3243031684378366 km         Vel. Apo:         7.722032341312458 km/sec
 Mean Mot.:        0.0663147949367717 deg/sec
    Period:         5428.652841997695 sec          Period:         90.47754736662824 min
    Period:         1.507959122777137 hr           Period:       0.06283163011571406 day
              Time Past Periapsis:          1032.054132491786 sec
          Time Past Ascending Node:         5235.111483469278 sec
   Beta Angle (Orbit plane to Sun):           5.25382696459117 deg
Mean Sidereal Greenwich Hour Angle:           207.981015646054 deg

Geodetic Parameters:
  Latitude:        -5.982655769958551 deg
 Longitude:         136.7898931479016 deg
  Altitude:      296.5992134540193206 km
Geocentric Parameters:
  Latitude:        -5.944660143338384 deg
 Longitude:         136.7898931479016 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     5428.6528419976939404 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 08:39:29.278  UTC Julian Date: 2461021.86075553
Julian Ephemeris Date: 2461021.86155627
Time past epoch: 49169.3 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5521.2188438793637033 km              Vx:       -4.3531723860735188 km/sec
         Y:     3286.7278983663049985 km              Vy:        5.9093079260466910 km/sec
         Z:     1995.9619958170662812 km              Vz:        3.0116123885371158 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2316426327515728 km            RAAN:         355.7161055868472 deg
       ecc:        0.0676721022413488                  w:         14.29383366262891 deg
       inc:         28.40963215253119 deg             TA:         24.27941113940199 deg

Parameter Set Type:  Spherical
 Right Asc:         30.76492712714889 deg     Horiz. FPA:         1.501328580715513 deg
      Decl:         17.25655495254434 deg        Azimuth:         67.07679494905803 deg
       |R|:     6728.3208966765250807 km             |V|:        7.9334632517269217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.73118079120329 deg       Mean Anom:         21.23295091994038 deg
 Long Peri:         10.00993924947607 deg        Arg. Lat:          38.5732448020309 deg
 True Long:         34.28935038887806 deg        Vert FPA:         88.49867141928449 deg
  Ang. Mom:         53360.56255334187 km^2/sec          p:     7143.3679935076306720 km
        C3:        -55.54453386537631 km^2/sec^2   Energy:        -27.77226693268815 km^2/sec^2
   Vel. RA:         126.3777247022548 deg       Vel. Decl:         22.30945234306578 deg
 Rad. Peri:     6690.6009612049056159 km        Vel. Peri:        7.9754513627027315 km/sec
  Rad. Apo:     7661.8623240605993487 km         Vel. Apo:         6.964437664949072 km/sec
 Mean Mot.:       0.05950409395239558 deg/sec
    Period:         6050.003891967616 sec          Period:         100.8333981994603 min
    Period:         1.680556636657671 hr           Period:       0.07002319319406963 day
              Time Past Periapsis:          356.8317658433241 sec
          Time Past Ascending Node:         566.3905527257348 sec
   Beta Angle (Orbit plane to Sun):           5.17972731078544 deg
Mean Sidereal Greenwich Hour Angle:           211.175479876071 deg

Geodetic Parameters:
  Latitude:         17.48628779704542 deg
 Longitude:         179.9445257274535 deg
  Altitude:      352.1005708629554647 km
Geocentric Parameters:
  Latitude:         17.38205222681539 deg
 Longitude:         179.9445257274535 deg
```

```
Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.058 kg
     Total Mass:  1370.06 kg
Area/Mass Ratio:  1.45979e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     6050.0038919676153455 sec


***<<<< End of Sequence: Target Sequence >>>>***


---------------------------------
Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 08:39:29.278  UTC Julian Date: 2461021.86075553
Julian Ephemeris Date: 2461021.86155627
Time past epoch: 49169.3 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5521.2188438793637033 km             Vx:       -4.3531723860735188 km/sec
         Y:     3286.7278983663049985 km             Vy:        5.9093079260466910 km/sec
         Z:     1995.9619958170662812 km             Vz:        3.0116123885371158 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2316426327515728 km           RAAN:         355.7161055868472 deg
       ecc:        0.0676721022413488                 w:         14.29383366262891 deg
       inc:         28.40963215253119 deg            TA:         24.27941113940199 deg

Parameter Set Type:  Spherical
 Right Asc:         30.76492712714889 deg     Horiz. FPA:         1.501328580715513 deg
      Decl:         17.25655495254434 deg        Azimuth:         67.07679494905803 deg
       |R|:     6728.3208966765250807 km            |V|:        7.9334632517269217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.73118079120329 deg       Mean Anom:         21.23295091994038 deg
 Long Peri:         10.00993924947607 deg        Arg. Lat:          38.5732448020309 deg
 True Long:         34.28935038887806 deg        Vert FPA:         88.49867141928449 deg
  Ang. Mom:         53360.56255334187 km^2/sec          p:     7143.3679935076306720 km
        C3:        -55.54453386537631 km^2/sec^2   Energy:        -27.77226693268815 km^2/sec^2
   Vel. RA:         126.3777247022548 deg       Vel. Decl:         22.30945234306578 deg
 Rad. Peri:     6690.6009612049056159 km        Vel. Peri:        7.9754513627027315 km/sec
  Rad. Apo:     7661.8623240605993487 km         Vel. Apo:         6.964437664949072 km/sec
 Mean Mot.:       0.05950409395239558 deg/sec
    Period:         6050.003891967616 sec          Period:         100.8333981994603 min
    Period:         1.680556636657671 hr           Period:       0.07002319319406963 day
             Time Past Periapsis:           356.8317658433241 sec
          Time Past Ascending Node:         566.3905527257348 sec
   Beta Angle (Orbit plane to Sun):          5.17972731078544 deg
Mean Sidereal Greenwich Hour Angle:          211.175479876071 deg

Geodetic Parameters:
  Latitude:         17.48628779704542 deg
 Longitude:         179.9445257274535 deg
  Altitude:      352.1005708629554647 km
Geocentric Parameters:
  Latitude:         17.38205222681539 deg
 Longitude:         179.9445257274535 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.058 kg
     Total Mass:  1370.06 kg
Area/Mass Ratio:  1.45979e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

```
***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate1
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  19 Dec 2025 08:39:29.278 [2461028.86075553]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    8189
  Average step size:  73.8522 sec
  Largest step size:  83.176 sec
  Smallest step size:  44.375 sec

----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 19 Dec 2025 08:39:29.278  UTC Julian Date: 2461028.86075553
Julian Ephemeris Date: 2461028.86155627
Time past epoch: 653969 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -2163.3347077274434014 km              Vx:       -6.9371981624747914 km/sec
         Y:     6439.6400751213468538 km              Vy:       -1.0600658484548673 km/sec
         Z:     1592.2659480540271488 km              Vz:       -3.0721564297042168 km/sec

Parameter Set Type:  Keplerian
       sma:     7173.2135470370722032 km            RAAN:         314.3536937715841 deg
       ecc:        0.0672995221158185                  w:         81.59603322820132 deg
       inc:         28.31751949586567 deg             TA:         69.64813601293164 deg

Parameter Set Type:  Spherical
 Right Asc:         108.5692756017059 deg     Horiz. FPA:         3.528118456874083 deg
      Decl:         13.19129257405914 deg        Azimuth:         115.2854976396858 deg
       |R|:     6977.4130022594645197 km             |V|:        7.6607181828518476 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         66.07184147712924 deg       Mean Anom:         62.54726599997741 deg
 Long Peri:         35.94972699978537 deg        Arg. Lat:          151.244169241133 deg
 True Long:          105.597863012717 deg        Vert FPA:         86.47188154312592 deg
  Ang. Mom:         53350.68805581993 km^2/sec          p:     7140.7244440531012515 km
        C3:        -55.56790396469427 km^2/sec^2   Energy:        -27.78395198234713 km^2/sec^2
   Vel. RA:         188.6880989148032 deg       Vel. Decl:         -23.6424099740614 deg
 Rad. Peri:     6690.4597032867623057 km        Vel. Peri:        7.9741438438992196 km/sec
  Rad. Apo:     7655.9673907873821008 km         Vel. Apo:         6.968510357034457 km/sec
 Mean Mot.:       0.05954165200047755 deg/sec
    Period:         6046.187633442093 sec          Period:         100.7697938907015 min
    Period:         1.679496564845026 hr           Period:       0.06997902353520941 day
               Time Past Periapsis:          1050.479183874101 sec
          Time Past Ascending Node:          2293.783240432894 sec
   Beta Angle (Orbit plane to Sun):          -1.71945816629326 deg
Mean Sidereal Greenwich Hour Angle:           218.075011478669 deg

Geodetic Parameters:
  Latitude:         13.22486774234803 deg
 Longitude:        -109.1407254134714 deg
  Altitude:      600.3869580943985511 km
Geocentric Parameters:
  Latitude:         13.14676953385449 deg
 Longitude:        -109.1407254134714 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  870.058 kg
     Total Mass:  1370.06 kg
Area/Mass Ratio:  1.45979e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


                              Valk4 - Astrogator Mission Control Sequence Summary
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***-------------------------------------------------------------------------***
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle
```

```
Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
             Time Past Periapsis:            1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
```

```
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
             Time Past Periapsis:           1159.77625053902 sec
         Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                     Pa
   Fuel Density:  1000                     kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 12:20:13.244 [2461022.01404218]:  Stopped on: Periapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    698
  Average step size:  66.3056 sec
  Largest step size:  68.467 sec
  Smallest step size:  41.067 sec

----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 12 Dec 2025 12:20:13.244  UTC Julian Date: 2461022.01404218
Julian Ephemeris Date: 2461022.01484292
Time past epoch: 62413.2 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -6435.2388504077744074 km             Vx:        1.9627103390121479 km/sec
         Y:    -1348.4953865265035802 km             Vy:       -6.6342871847076088 km/sec
         Z:    -1057.9561547289024475 km             Vz:       -3.4823774648890202 km/sec

Parameter Set Type:  Keplerian
       sma:     6675.1115038002108122 km           RAAN:         354.5810541410926 deg
       ecc:        0.0023267296107671                 w:         199.4611600175548 deg
       inc:         28.47925227436462 deg            TA:         359.9992859852982 deg

Parameter Set Type:  Spherical
 Right Asc:         191.8350067641019 deg     Horiz. FPA:    -1.657462690823182e-06 deg
      Decl:        -9.140865415674448 deg        Azimuth:         117.0895429676731 deg
       |R|:     6659.5803242103484081 km             |V|:        7.7455116765780989 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         359.9992876447091 deg       Mean Anom:         359.9992893021673 deg
```

```
 Long Peri:         194.0422141586474 deg        Arg. Lat:          199.460446002853 deg
 True Long:         194.0415001439456 deg        Vert FPA:         90.00000165746269 deg
  Ang. Mom:         51581.85716228099 km^2/sec          p:     6675.0753669447676657 km
        C3:        -59.71442443666635 km^2/sec^2   Energy:        -29.85721221833317 km^2/sec^2
   Vel. RA:         286.4805057203778 deg       Vel. Decl:        -26.71798529294295 deg
 Rad. Peri:     6659.5803242091460561 km        Vel. Peri:        7.7455116765794934 km/sec
  Rad. Apo:     6690.6426833912755683 km         Vel. Apo:         7.709551922467301 km/sec
 Mean Mot.:        0.0663290432597823 deg/sec
    Period:         5427.486698248233 sec          Period:         90.45811163747054 min
    Period:         1.507635193957842 hr           Period:       0.06281813308157676 day
              Time Past Periapsis:           5427.475983517584 sec
          Time Past Ascending Node:          3008.477376709841 sec
   Beta Angle (Orbit plane to Sun):           5.29487334209913 deg
Mean Sidereal Greenwich Hour Angle:           266.509760740163 deg

Geodetic Parameters:
  Latitude:        -9.341961629378483 deg
 Longitude:        -74.33777015833071 deg
  Altitude:      282.0023764144594907 km
Geocentric Parameters:
  Latitude:        -9.283116666535506 deg
 Longitude:        -74.33777015833071 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: TargeterSequence
Name: Target Sequence
User Comment:  Sequence that runs targeting profiles

Sequence Start: 12 Dec 2025 12:20:13.244 UTCG;   2461022.01404218 UTC Julian Date
Sequence Stop:  12 Dec 2025 13:10:13.151 UTCG;   2461022.04876332 UTC Julian Date

***<<<< Start of Sequence: Target Sequence >>>>***


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Target Sequence.Propagate
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 13:03:50.263 [2461022.04433175]:  Stopped on: Apoapsis; Run Sequence STOP

Propagation Statistics:
  Number of steps:    40
  Average step size:  65.9348 sec
  Largest step size:  67.015 sec
  Smallest step size:  58.014 sec

----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 12 Dec 2025 13:03:50.263  UTC Julian Date: 2461022.04433175
Julian Ephemeris Date: 2461022.04513249
Time past epoch: 65030.3 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6590.5717766956913692 km              Vx:       -1.1661626916401360 km/sec
         Y:      740.0401787369560225 km              Vy:        6.7484772779626345 km/sec
         Z:      751.6145051101290164 km              Vz:        3.5810040802912826 km/sec

Parameter Set Type:  Keplerian
       sma:     6675.0138336518657525 km            RAAN:         354.3486746904143 deg
       ecc:        0.0000851721526855                  w:         13.65938651960293 deg
       inc:         28.48015655021798 deg             TA:     0.0002351893333769571 deg

Parameter Set Type:  Spherical
 Right Asc:         6.406774855179501 deg     Horiz. FPA:     2.002987388420337e-08 deg
      Decl:         6.465837706689493 deg        Azimuth:         62.20346262816211 deg
       |R|:     6674.4453083544531182 km             |V|:        7.7282256189641716 km/sec
```

```
Other Elliptic Orbit Parameters :
 Ecc. Anom:     0.0002351677625431845 deg       Mean Anom:     0.0002351477327986065 deg
 Long Peri:         8.008061210017269 deg        Arg. Lat:          13.6596217089363 deg
 True Long:         8.008296399350639 deg        Vert FPA:         89.99999997997013 deg
  Ang. Mom:          51581.6192244001 km^2/sec          p:     6675.0137852293419201 km
        C3:        -59.71529819016535 km^2/sec^2   Energy:        -29.85764909508267 km^2/sec^2
   Vel. RA:         99.80410771055787 deg       Vel. Decl:         27.60458283808357 deg
 Rad. Peri:     6674.4453083544485708 km        Vel. Peri:        7.7282256189641769 km/sec
  Rad. Apo:     6675.5823589492838437 km         Vel. Apo:          7.72690927185548 km/sec
 Mean Mot.:       0.06633049907501078 deg/sec
    Period:         5427.367576307378 sec          Period:         90.45612627178963 min
    Period:         1.507602104529827 hr           Period:       0.06281675435540947 day
              Time Past Periapsis:        0.003545092168425967 sec
          Time Past Ascending Node:           205.8979983605407 sec
   Beta Angle (Orbit plane to Sun):            5.30362266535682 deg
Mean Sidereal Greenwich Hour Angle:            277.443861400546 deg

Geodetic Parameters:
  Latitude:         6.652266335984836 deg
 Longitude:         89.29701441165172 deg
  Altitude:      296.5930006912603289 km
Geocentric Parameters:
  Latitude:         6.610090056528456 deg
 Longitude:         89.29701441165172 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-----------------------------------------------------------------------------***
MCS Segment Type: Maneuver:Finite
Name: Target Sequence.Maneuver
User Comment:  Maneuvers satellite with an impulsive burn or finite burn

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  12 Dec 2025 13:10:13.151 [2461022.04876332]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    11
  Average step size:  64.9054 sec
  Largest step size:  66.475 sec
  Smallest step size:  60 sec

Maneuver Summary:
  Maneuver Start: 12 Dec 2025 12:57:27.376 UTCG;   2461022.03990019 UTC Julian Date
  Maneuver Stop:  12 Dec 2025 13:10:13.151 UTCG;   2461022.04876332 UTC Julian Date
  Duration:  765.774 sec
  Fuel Used:        130.1453806681793 kg
  DeltaV Magnitude:        267.0169112344254 m/sec
  Maneuver Direction Specification:  Thrust Vector
  Maneuver direction is updated during maneuver.
  Burn centering is ON.
  Thrust Efficiency:  1  (Affects acceleration, but not mass flow rate)

Thrust vector at maneuver start with respect to VNC(Earth) axes:

       X (Velocity):        0.9987521608999563
         Y (Normal):      -0.03470869287013872
      Z (Co-Normal):       -0.0359086025475028
            Azimuth:        -1.990345251576628 deg
          Elevation:        -2.057853778149658 deg

Thrust vector at maneuver start with respect to Earth Inertial axes:

              X:        0.2547572772179428
              Y:        0.8614938961240182
              Z:         0.439234557663169
        Azimuth:         73.52622401222224 deg
      Elevation:          26.0550531030021 deg

Thrust vector at maneuver stop with respect to VNC(Earth) axes:

       X (Velocity):        0.9987521608999563
         Y (Normal):      -0.03470869287013878
```

```
      Z (Co-Normal):      -0.03590860254750286
            Azimuth:        -1.990345251576631 deg
          Elevation:        -2.057853778149661 deg

Thrust vector at maneuver stop with respect to Earth Inertial axes:

              X:       -0.5751654045329948
              Y:        0.7476473133484476
              Z:        0.3319762826938766
        Azimuth:         127.5710997049192 deg
      Elevation:         19.38877177837035 deg

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to VNC(Earth) axes:

       X (Velocity):         229.0235450617326 m/sec
         Y (Normal):        -9.430435020665257 m/sec
      Z (Co-Normal):        -119.7956188479853 m/sec
            Azimuth:        -2.357919395672322 deg
          Elevation:        -27.59284267619613 deg
          Magnitude:          258.634312480951 m/sec

Integrated Inertial DeltaV vector referenced to maneuver start thrust vector with respect to Earth Inertial axes:

              X:        -48.22430604783533 m/sec
              Y:          228.917262601034 m/sec
              Z:         110.2860407392893 m/sec
        Azimuth:           101.8961350261 deg
      Elevation:         25.24038131663476 deg
      Magnitude:          258.634312480951 m/sec

Thrust vector at maneuver start with respect to spacecraft body axes:

              X:                         1
              Y:                         0
              Z:    -1.318389841742373e-16
        Azimuth:                         0 deg
      Elevation:                         0 deg

Thrust vector at maneuver stop with respect to spacecraft body axes:

              X:                         1
              Y:    -1.110223024625157e-16
              Z:     1.804112415015879e-16
        Azimuth:    -6.361109362927032e-15 deg
      Elevation:                         0 deg

Attitude with respect to Earth Inertial axes:
  ---  ----Maneuver Start-----  -----Maneuver Stop-----
  qx:       0.6754063806819626       0.3864069116949996
  qy:       0.3188798332250585       0.4837176113574437
  qz:       0.5205161028761622       0.7440429925274121
  qs:       0.4137690896418718       0.2512110593262871

Data for Engine "Constant_Thrust_and_Isp":
  UserComment: Engine that has a constant Thrust and Isp
  Description: Engine that has a constant Thrust and Isp

  Engine values at beginning of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

  Engine values at end of segment:
            Thrust:                      500 N
               Isp:                      300 s
    Mass Flow Rate:      -0.1699527021629881 kg/sec

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 12 Dec 2025 12:57:27.376  UTC Julian Date: 2461022.03990019
Julian Ephemeris Date: 2461022.04070093
Time past epoch: 64647.4 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6384.5676132655989932 km              Vx:        2.2247464758210280 km/sec
         Y:    -1831.5602711747887952 km              Vy:        6.4631606149785510 km/sec
         Z:     -647.8176945039881502 km              Vz:        3.6082490571565971 km/sec

Parameter Set Type:  Keplerian
       sma:     6675.1107233323064065 km            RAAN:          354.349803039327 deg
       ecc:        0.0006038518058837                 w:         280.1799483268379 deg
       inc:          28.4811089780124 deg            TA:         68.07481151530386 deg

Parameter Set Type:  Spherical
 Right Asc:         343.9932422785686 deg     Horiz. FPA:       0.03208851227868336 deg
      Decl:        -5.570567694421731 deg        Azimuth:         62.02411806499071 deg
       |R|:     6673.6035543489033444 km            |V|:        7.7292563209638399 km/sec
```

```
Other Elliptic Orbit Parameters :
 Ecc. Anom:          68.0427193804027 deg       Mean Anom:         68.01063087061942 deg
 Long Peri:         274.5297513661648 deg        Arg. Lat:         348.2547598421417 deg
 True Long:         342.6045628814687 deg        Vert FPA:         89.96791148772131 deg
  Ang. Mom:         51581.98436655192 km^2/sec          p:     6675.1082893399334353 km
        C3:        -59.71443141860173 km^2/sec^2   Energy:        -29.85721570930086 km^2/sec^2
   Vel. RA:         71.00553360011945 deg       Vel. Decl:         27.82872369483784 deg
 Rad. Peri:     6671.0799456675476904 km        Vel. Peri:        7.7321790154907699 km/sec
  Rad. Apo:     6679.1415009970642132 km         Vel. Apo:         7.722846470441111 km/sec
 Mean Mot.:       0.06632905489277829 deg/sec
    Period:          5427.48574635873 sec          Period:         90.45809577264549 min
    Period:         1.507634929544092 hr           Period:       0.06281812206433715 day
              Time Past Periapsis:          1025.352026808755 sec
          Time Past Ascending Node:         5250.469516014547 sec
   Beta Angle (Orbit plane to Sun):           5.30467541590794 deg
Mean Sidereal Greenwich Hour Angle:           275.844130775165 deg

Geodetic Parameters:
  Latitude:        -5.466521481202208 deg
 Longitude:         68.48559619083736 deg
  Altitude:      295.6590749453582703 km
Geocentric Parameters:
  Latitude:        -5.431757514068017 deg
 Longitude:         68.48559619083736 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     5427.4857463587295570 sec

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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 13:10:13.151  UTC Julian Date: 2461022.04876332
Julian Ephemeris Date: 2461022.04956406
Time past epoch: 65413.2 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5521.9890007941830845 km              Vx:       -4.3438483288293863 km/sec
         Y:     3252.6619941285644018 km              Vy:        5.9411883310183118 km/sec
         Z:     2046.8834591068603004 km              Vz:        2.9637907241428856 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2267913357445650 km            RAAN:         354.3006965385188 deg
       ecc:        0.0677950825984662                  w:         15.41608791327034 deg
       inc:         28.40425987392738 deg             TA:         24.34531667681899 deg

Parameter Set Type:  Spherical
 Right Asc:          30.4997064515103 deg     Horiz. FPA:         1.507774719597173 deg
      Decl:          17.7128842241356 deg        Azimuth:         67.42654937591031 deg
       |R|:     6727.6968174930134410 km             |V|:        7.9341535493723221 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.79037350942329 deg       Mean Anom:         21.28572043091325 deg
 Long Peri:         9.716784451789204 deg        Arg. Lat:         39.76140459008932 deg
 True Long:         34.06210112860821 deg        Vert FPA:         88.49222528040282 deg
  Ang. Mom:         53360.09798201162 km^2/sec          p:     7143.2436099042315618 km
        C3:        -55.54457141477918 km^2/sec^2   Energy:        -27.77228570738959 km^2/sec^2
   Vel. RA:         126.1720808232329 deg       Vel. Decl:         21.93462628183552 deg
 Rad. Peri:     6689.7139032718123417 km        Vel. Peri:        7.9764394641621683 km/sec
  Rad. Apo:     7662.7396793996795168 km         Vel. Apo:         6.963579635292008 km/sec
 Mean Mot.:       0.05950415429164281 deg/sec
    Period:         6049.997757056787 sec          Period:         100.8332959509465 min
    Period:         1.680554932515774 hr           Period:       0.07002312218815726 day
              Time Past Periapsis:          357.7182246232306 sec
          Time Past Ascending Node:         583.7210470634969 sec
   Beta Angle (Orbit plane to Sun):           5.22956248799732 deg
Mean Sidereal Greenwich Hour Angle:           279.043592194025 deg

Geodetic Parameters:
  Latitude:         17.94533837777185 deg
 Longitude:         111.811649163771 deg
```

```
  Altitude:      351.5751673933114034 km
Geocentric Parameters:
  Latitude:         17.83871693173932 deg
 Longitude:         111.8116491637771 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  869.855 kg
     Total Mass:  1369.85 kg
Area/Mass Ratio:  1.46001e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


User-selected results:
    Orbit Period =     6049.9977570567862131 sec


***<<<< End of Sequence: Target Sequence >>>>***


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Satellite State at End of Segment:
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UTC Gregorian Date: 12 Dec 2025 13:10:13.151  UTC Julian Date: 2461022.04876332
Julian Ephemeris Date: 2461022.04956406
Time past epoch: 65413.2 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     5521.9890007941830845 km             Vx:       -4.3438483288293863 km/sec
         Y:     3252.6619941285644018 km             Vy:        5.9411883310183118 km/sec
         Z:     2046.8834591068603004 km             Vz:        2.9637907241428856 km/sec

Parameter Set Type:  Keplerian
       sma:     7176.2267913357445650 km           RAAN:         354.3006965385188 deg
       ecc:        0.0677950825984662                w:         15.41608791327034 deg
       inc:         28.40425987392738 deg            TA:         24.34531667681899 deg

Parameter Set Type:  Spherical
 Right Asc:          30.4997064515103 deg     Horiz. FPA:         1.507774719597173 deg
      Decl:          17.7128842241356 deg        Azimuth:         67.42654937591031 deg
       |R|:     6727.6968174930134410 km            |V|:        7.9341535493723221 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         22.79037350942329 deg       Mean Anom:         21.28572043091325 deg
 Long Peri:         9.716784451789204 deg        Arg. Lat:         39.76140459008932 deg
 True Long:         34.06210112860821 deg        Vert FPA:         88.49222528040282 deg
  Ang. Mom:         53360.09798201162 km^2/sec          p:     7143.2436099042315618 km
        C3:        -55.54457141477918 km^2/sec^2   Energy:        -27.77228570738959 km^2/sec^2
   Vel. RA:         126.1720808232329 deg       Vel. Decl:         21.93462628183552 deg
 Rad. Peri:     6689.7139032718123417 km        Vel. Peri:        7.9764394641621683 km/sec
  Rad. Apo:     7662.7396793996795168 km         Vel. Apo:         6.963579635292008 km/sec
 Mean Mot.:       0.05950415429164281 deg/sec
    Period:         6049.997757056787 sec          Period:         100.8332959509465 min
    Period:         1.680554932515774 hr           Period:       0.07002312218815726 day
              Time Past Periapsis:          357.7182246232306 sec
          Time Past Ascending Node:         583.7210470634969 sec
   Beta Angle (Orbit plane to Sun):           5.22956248799732 deg
Mean Sidereal Greenwich Hour Angle:           279.043592194025 deg

Geodetic Parameters:
  Latitude:         17.94533837777185 deg
 Longitude:         111.8116491637771 deg
  Altitude:      351.5751673933114034 km
Geocentric Parameters:
  Latitude:         17.83871693173932 deg
 Longitude:         111.8116491637771 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  869.855 kg
     Total Mass:  1369.85 kg
Area/Mass Ratio:  1.46001e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
```

```
            Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


***-------------------------------------------------------------------------***
MCS Segment Type: Propagate
Name: Propagate1
User Comment:  Propagates until stopping conditions are met

Propagator model used: Earth_HPOP_Default_v10  (Default Earth HPOP settings for STK 10.0)
Integrator used: RKF7th8th  (Runge-Kutta-Fehlberg 7(8))

Stopping Condition Information (Gregorian Date [Julian Date]):
  19 Dec 2025 13:10:13.151 [2461029.04876332]:  Stopped on: Duration; Run Sequence STOP

Propagation Statistics:
  Number of steps:    8183
  Average step size:  73.9037 sec
  Largest step size:  83.211 sec
  Smallest step size:  47.069 sec

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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 19 Dec 2025 13:10:13.151  UTC Julian Date: 2461029.04876332
Julian Ephemeris Date: 2461029.04956406
Time past epoch: 670213 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:    -2065.3978644913190692 km             Vx:       -6.9466504066342010 km/sec
         Y:     6474.7230325620757867 km             Vy:       -0.9630969535152871 km/sec
         Z:     1561.5039864674608907 km             Vz:       -3.0939850762371219 km/sec

Parameter Set Type:  Keplerian
       sma:     7173.3971220276607710 km           RAAN:         312.9323297423092 deg
       ecc:        0.0673966277977830                 w:         82.71898727371511 deg
       inc:         28.31709652331259 deg            TA:          69.1121380601255 deg

Parameter Set Type:  Spherical
 Right Asc:         107.6923713752376 deg     Horiz. FPA:         3.518672891380564 deg
      Decl:         12.93983428994925 deg        Azimuth:         115.4082107444134 deg
       |R|:     6973.2489835649012093 km            |V|:        7.6652626351508379 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         65.54435215025454 deg       Mean Anom:         62.02925964547062 deg
 Long Peri:         35.65131701602431 deg        Arg. Lat:         151.8311253338406 deg
 True Long:         104.7634550761499 deg        Vert FPA:         86.48132710861944 deg
  Ang. Mom:         53351.02021958333 km^2/sec          p:     7140.8133612676629127 km
        C3:        -55.56648192193352 km^2/sec^2   Energy:        -27.78324096096676 km^2/sec^2
   Vel. RA:         187.8932797846895 deg       Vel. Decl:        -23.80575465415891 deg
 Rad. Peri:     6689.9343461486741944 km        Vel. Peri:        7.9748197006293422 km/sec
  Rad. Apo:     7656.8598979066473476 km         Vel. Apo:         6.967741467252035 km/sec
 Mean Mot.:       0.05953936641219432 deg/sec
    Period:         6046.419733587692 sec          Period:         100.7736622264615 min
    Period:         1.679561037107692 hr           Period:       0.06998170987948717 day
              Time Past Periapsis:            1041.81927661169 sec
          Time Past Ascending Node:           2303.381793076949 sec
   Beta Angle (Orbit plane to Sun):           -2.21242694497408 deg
Mean Sidereal Greenwich Hour Angle:            285.943123796623 deg

Geodetic Parameters:
  Latitude:          12.9741946783333 deg
 Longitude:        -177.8862174091562 deg
  Altitude:      596.1819188300860333 km
Geocentric Parameters:
  Latitude:         12.89742777101095 deg
 Longitude:        -177.8862174091562 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  869.855 kg
     Total Mass:  1369.85 kg
Area/Mass Ratio:  1.46001e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
            Cr:  1.000000
            Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

Valk1: Formatted STK MCS Summary

```
                                                                                    15 Dec 2025 10:41:30
Satellite-Valk1


                  Astrogator MCS Segment Summary Astrogator MCS Follow Summary
-----------------------------------------------------------------------------------------------------
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:     6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
             Time Past Periapsis:            1159.77625053902 sec
         Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


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Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial
```

```
Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523927 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232201796 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205355 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946475 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:     1.373554776971959e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:         76.90112810041059 deg       Mean Anom:         76.88407011249528 deg
 Long Peri:         281.6650460761578 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648821 km^2/sec          p:      6677.5867113024332866 km
        C3:        -59.69228427720376 km^2/sec^2   Energy:        -29.84614213860188 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871272991630 km        Vel. Peri:        7.7284401980431365 km/sec
  Rad. Apo:     6679.6284833191421058 km         Vel. Apo:         7.723716914095812 km/sec
 Mean Mot.:       0.06629215771297481 deg/sec
    Period:         5430.506600172711 sec          Period:         90.50844333621185 min
    Period:         1.508474055603531 hr           Period:       0.06285308565014712 day
               Time Past Periapsis:           1159.77625053902 sec
          Time Past Ascending Node:       3.865352482534945e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

Valk2: Formatted STK MCS Summary

```
                                                                                    15 Dec 2025 10:42:41
Satellite-Valk2


                  Astrogator MCS Segment Summary Astrogator MCS Follow Summary
-----------------------------------------------------------------------------------------------------
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:     6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
             Time Past Periapsis:          1159.776250539018 sec
         Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial
```

```
Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:      6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
               Time Past Periapsis:           1159.776250539018 sec
          Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

Valk3: Formatted STK MCS Summary

```
                                                                                    15 Dec 2025 10:42:54
Satellite-Valk3


                  Astrogator MCS Segment Summary Astrogator MCS Follow Summary
-----------------------------------------------------------------------------------------------------
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:     6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
             Time Past Periapsis:          1159.776250539018 sec
         Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial
```

```
Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:      6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
               Time Past Periapsis:          1159.776250539018 sec
          Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```

Valk4: Formatted STK MCS Summary

```
                                                                                    15 Dec 2025 10:43:14
Satellite-Valk4


                  Astrogator MCS Segment Summary Astrogator MCS Follow Summary
-----------------------------------------------------------------------------------------------------
MCS Segment Type: Follow
Name: Follow
User Comment:  Follows another vehicle

Stopping Condition Information (Gregorian Date [Julian Date]):

----------------------------------------
Satellite State at Beginning of Segment:
----------------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial

Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:     6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
             Time Past Periapsis:          1159.776250539018 sec
         Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):           5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:           73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000


----------------------------------
Satellite State at End of Segment:
----------------------------------

UTC Gregorian Date: 11 Dec 2025 23:28:57.781  UTC Julian Date: 2461021.47844654
Julian Ephemeris Date: 2461021.47924729
Time past epoch: 16137.8 sec   (Epoch in UTC Gregorian Date: 11 Dec 2025 19:00:00.000)

State Vector in Coordinate System: Earth Inertial
```

```
Parameter Set Type:  Cartesian
         X:     6675.0835287679083194 km              Vx:        0.1701940855637216 km/sec
         Y:     -165.0901289838727166 km              Vy:        6.7884131218611676 km/sec
         Z:        0.0000000000160071 km              Vz:        3.6863310315523932 km/sec

Parameter Set Type:  Keplerian
       sma:     6677.5873352232229081 km            RAAN:         358.5832327556222 deg
       ecc:        0.0003056714938274                  w:         283.0818133205357 deg
       inc:         28.49586471175197 deg             TA:         76.91818667946455 deg

Parameter Set Type:  Spherical
 Right Asc:         358.5832327556225 deg     Horiz. FPA:       0.01705798820822765 deg
      Decl:      1.37355477697196e-13 deg        Azimuth:         61.50413528824804 deg
       |R|:     6677.1247454811373245 km             |V|:        7.7266134375937217 km/sec


Other Elliptic Orbit Parameters :
 Ecc. Anom:          76.9011281004104 deg       Mean Anom:         76.88407011249508 deg
 Long Peri:          281.665046076158 deg        Arg. Lat:     2.544443745170813e-13 deg
 True Long:         358.5832327556226 deg        Vert FPA:         89.98294201179176 deg
  Ang. Mom:         51591.55949648822 km^2/sec          p:     6677.5867113024360151 km
        C3:        -59.69228427720373 km^2/sec^2   Energy:        -29.84614213860187 km^2/sec^2
   Vel. RA:         88.56382334788049 deg       Vel. Decl:         28.49586333329686 deg
 Rad. Peri:     6675.5461871273028009 km        Vel. Peri:        7.7284401980431348 km/sec
  Rad. Apo:     6679.6284833191448342 km         Vel. Apo:          7.72371691409581 km/sec
 Mean Mot.:       0.06629215771297477 deg/sec
    Period:         5430.506600172715 sec          Period:         90.50844333621191 min
    Period:         1.508474055603532 hr           Period:       0.06285308565014716 day
               Time Past Periapsis:          1159.776250539018 sec
          Time Past Ascending Node:       4.092726157978177e-12 sec
   Beta Angle (Orbit plane to Sun):            5.09248932109083 deg
Mean Sidereal Greenwich Hour Angle:            73.1674230549042 deg

Geodetic Parameters:
  Latitude:        0.1457840393089087 deg
 Longitude:        -74.25175553637644 deg
  Altitude:      298.9878828100316923 km
Geocentric Parameters:
  Latitude:        0.1448518098381939 deg
 Longitude:        -74.25175553637644 deg



Spacecraft Configuration:
      Drag Area:  20 m^2
       SRP Area:  20 m^2
       Dry Mass:  500 kg
      Fuel Mass:  1000 kg
     Total Mass:  1500 kg
Area/Mass Ratio:  1.33333e-08 km^2/kg
  Tank Pressure:  5000                      Pa
   Fuel Density:  1000                      kg/m^3
             Cr:  1.000000
             Cd:  2.200000
 Rad Press Area:  20 m^2
Rad Press Coeff:  1.000000
```